

DS7006 Quantitative Data Analysis

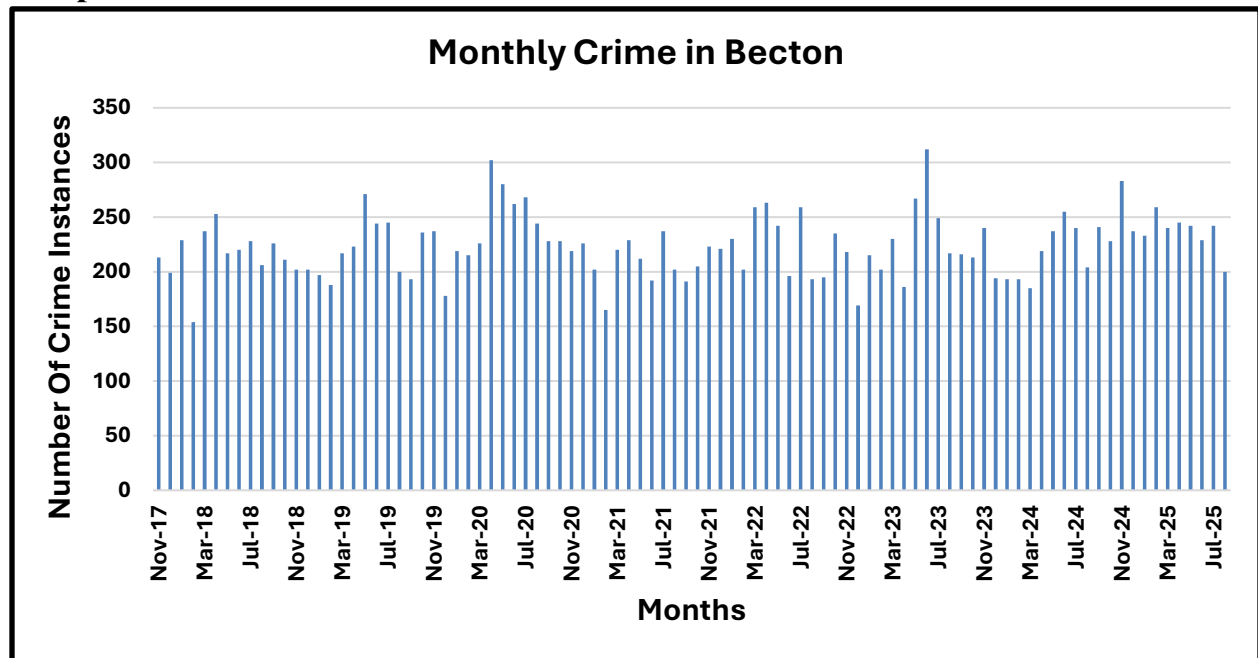
Portfolio
by
Abdul Hadi

Session 1

Trend, Distribution, Standardisation, Indexing

Task 1: (a) The time series graph from Task1 and your interpretation of it

Graph:



Interpretation

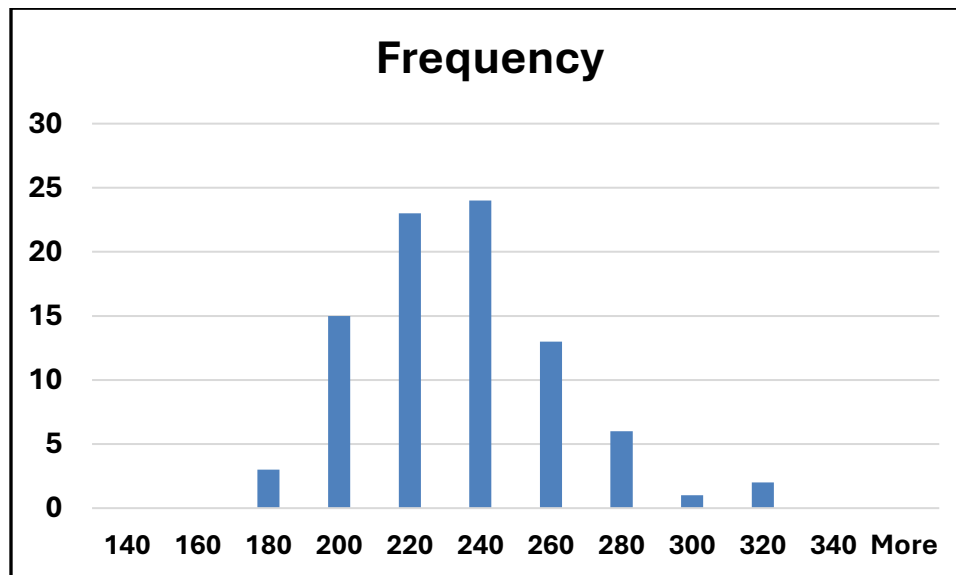
A time series column chart was created in Excel using monthly crime data for Beckton (2017–2025) to visualize changes over time. The data were plotted with months on the x-axis and total crime instances on the y-axis to identify patterns and fluctuations.

The chart shows that monthly crime levels generally range between 150 and 300 incidents. While the overall trend appears stable, noticeable peaks occur around 2020 and 2023, possibly reflecting external factors such as policing changes or Covid_19 restrictions.

Task b: the graphed distribution from Task 3 together with the values for the summary statistics

The monthly crime data for Beckton (2017–August 2025) shows a consistent pattern overall. The average number of crimes per month is about 224, with a median of 223, indicating balanced values without major outliers. The standard deviation of around 28 suggests moderate variation.

Graph:

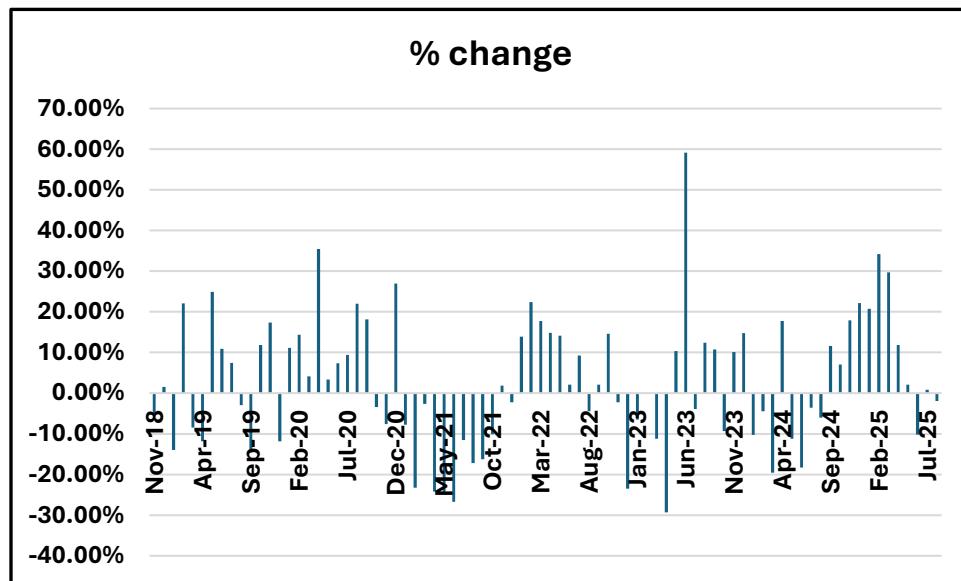


Interpretation

The histogram shows that most months record between 200 and 260 crimes, with only a few months exceeding 280. This slightly right-skewed distribution suggests that crime in Beckton is generally stable, with occasional short-term increases but no major long-term changes.

Task c: The second graph (percentage annual change) from Task 4 and interpretation of it

Graph:

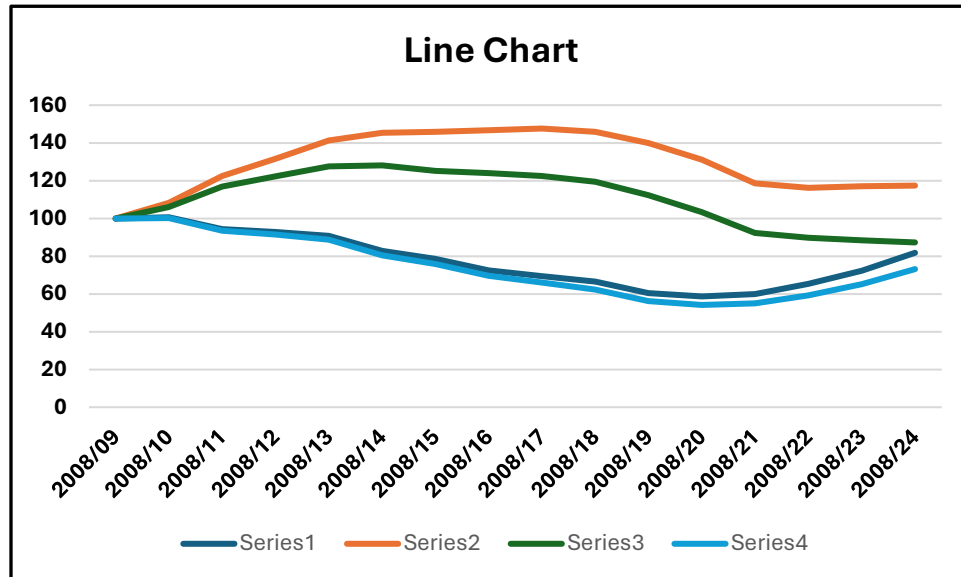


Interpretation

The percentage change chart shows how monthly crime levels in Beckton rose and fell between 2018 and 2025. The pattern reflects normal month to month variation, with several small increases and decreases throughout the period. Most changes are modest, though a few months, like mid of 2023, show a noticeable spike in crime compared to the previous month. Periods of decline are followed by short recoveries, suggesting no clear long-term rise or fall. Overall, the data indicates that crime in Beckton tends to fluctuate but remains relatively stable over time.

Task d: The line chart resulting from Task5 and after Task6. Write an interpretation of the line chart

Graph:

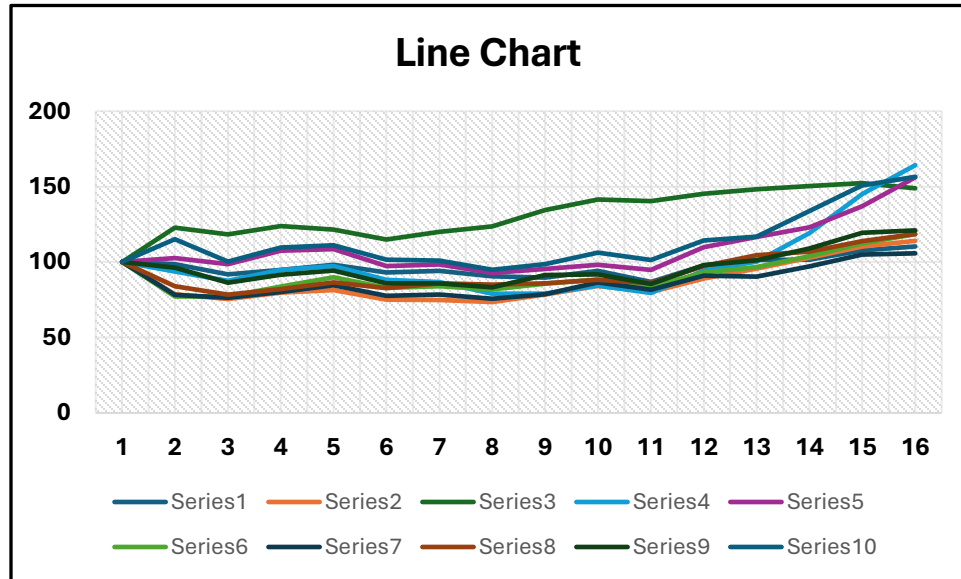


Interpretation:

The chart shows how four standardized data series changed over time, all starting from a base value of 100. Series 2 rose the most before gradually declining, while Series 3 followed a similar but smaller trend. Series 1 and 4 stayed relatively steady throughout. Overall, the graph highlights that some categories experienced stronger rises and falls, while others remained more stable.

Task e: Include the line graph from Task7 and again, write an interpretation

Graph:



Interpretation

The line chart compares changes in crime per 1,000 people across ten regions, indexed to 2007/08 = 100. Most regions show a small decline after the base year, followed by gradual recovery in later years. A few areas rise more sharply toward the end, indicating higher relative increases in crime. Overall, the chart suggests that while regional patterns differ, crime rates across most areas have stayed within a similar range over time.

Session 3

Database

Task 4: Construct a query to sort the SOIL_ID table for ph_val > 5 ordered by ph_val.

SQL Query:

```
SELECT soil_id, soil_type, ph_val, sur_id, assessed  
FROM SOIL_ID  
WHERE ph_val > 5  
ORDER BY ph_val;
```

Result

213	32	5.12	1	1991-12-24
222	26	5.13	3	1990-05-19
225	17	5.34	4	1992-07-04
141	17	5.43	3	1990-11-22
212	27	5.44	3	1992-07-04
245	21	5.74	4	1990-11-06
193	17	6.44	4	1990-05-19
170	30	6.54	4	1989-09-12
179	17	6.54	1	1992-09-17

Task 5: Construct a query to show the soil_id, ph-val and assessed for N.Brown in descending order of data assessed.

Sql Query:

```
SELECT s.soil_id, s.ph_val, s.assessed
FROM SOIL_ID AS s
JOIN SURVEYOR AS v ON s.sur_id = v.sur_id
WHERE v.surveyor = 'N.Brown'
ORDER BY s.assessed DESC;
```

Result:

261	3.29	1992-07-04
182	3.34	1991-12-24
254	4.32	1991-12-24
194	4.52	1990-11-06
216	4.73	1990-11-06
232	4.32	1990-05-19
274	4.56	1990-05-19
157	2.36	1989-07-12

Task 6: Construct a query to list ph_val and soil_name for series = 3 ordered by soil_name in alphabetical order.

Sql query:

```
SELECT t.soil_name, s.ph_val
FROM SOIL_ID AS s
JOIN SOIL_TYPE AS t ON s.soil_type = t.soil_type
WHERE t.series = 3
ORDER BY t.soil_name ASC;
```

Result

Clay	3.43
Clay	4.65
Clay	3.44
Clay	4.52
Clay	2.9
Glacial Sand	5.43
Glacial Sand	6.54
Glacial Sand	6.44
Glacial Sand	5.34
Glacial Sand	4.32
Gravel	2.36
Gravel	6.54
Gravel	3.54
Gravel	4.56
River Alluvium	3.34
River Alluvium	5.74
Sand	3.33
Sand	2.22

Reflection

This exercise improved my understanding of how relational databases are structured and how SQL can be used to manage and analyse data effectively. I learned how to design tables, define primary keys, import data, and write queries using filtering, sorting, and joins. These skills are valuable for my research and future studies, as they will help me handle large datasets and perform meaningful data analysis in a structured and accurate way.

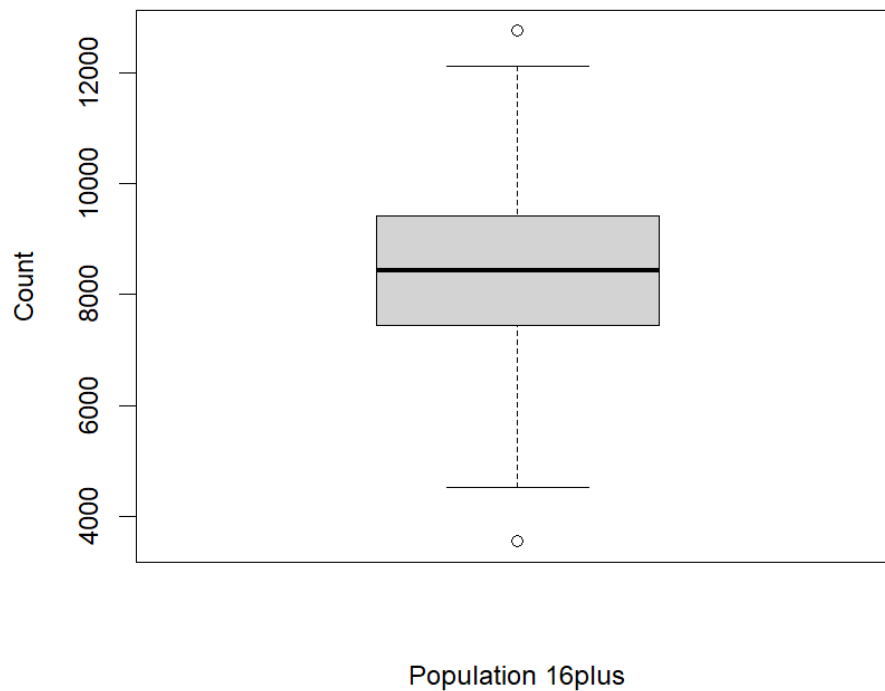
Session 4

Data Exploration and Graphics

Task 1

Section 2 : Create a simple boxplot of variable P16plus which is the population aged 16 and over for each Ward.

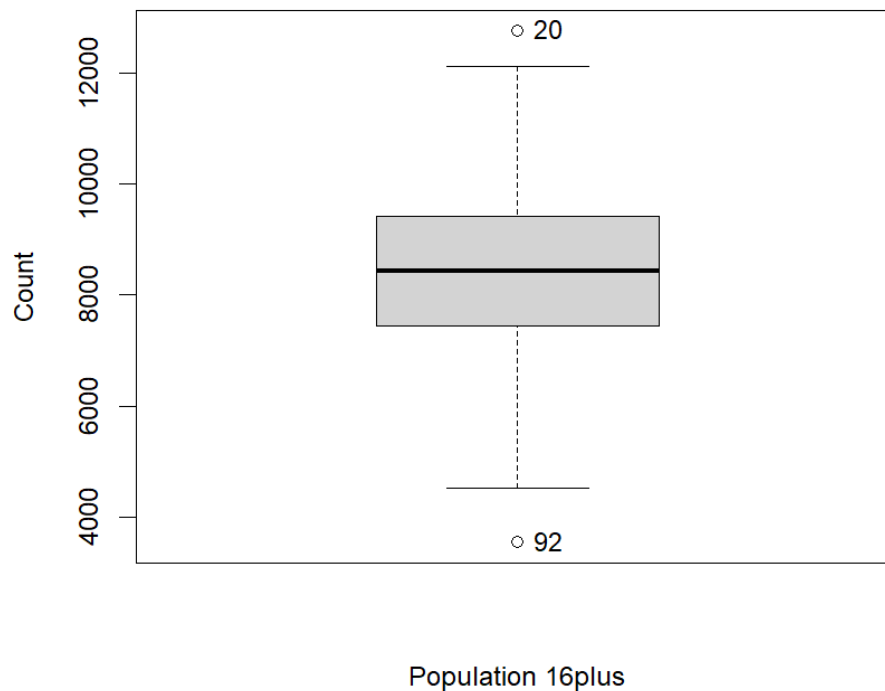
Boxplot:



Interpretation

The boxplot of P16plus shows that most wards in London have an adult population between 6,000 and 10,000, with a median around 8,500. The interquartile range (IQR) is relatively compact, suggesting that most wards have a similar population size. However, there are two visible outliers, one below and one above the main range. This indicates that a few wards have either significantly smaller or larger adult populations compared to the majority.

Second Boxplot (Outliers Labeled):



In the second boxplot, the outliers are labeled with their case numbers: 20 and 92.

Are there any outliers ?

Yes, there are two clear outliers in the distribution. One ward has a much higher population than the rest, while another has a noticeably lower population. These outliers stand out beyond the upper and lower whiskers of the boxplot.

Are they errors?

No, these outliers are not data errors. They represent genuine differences in ward population sizes. Such variation is expected in London, where some wards are more densely populated and others are more suburban or rural.

Which wards are they?

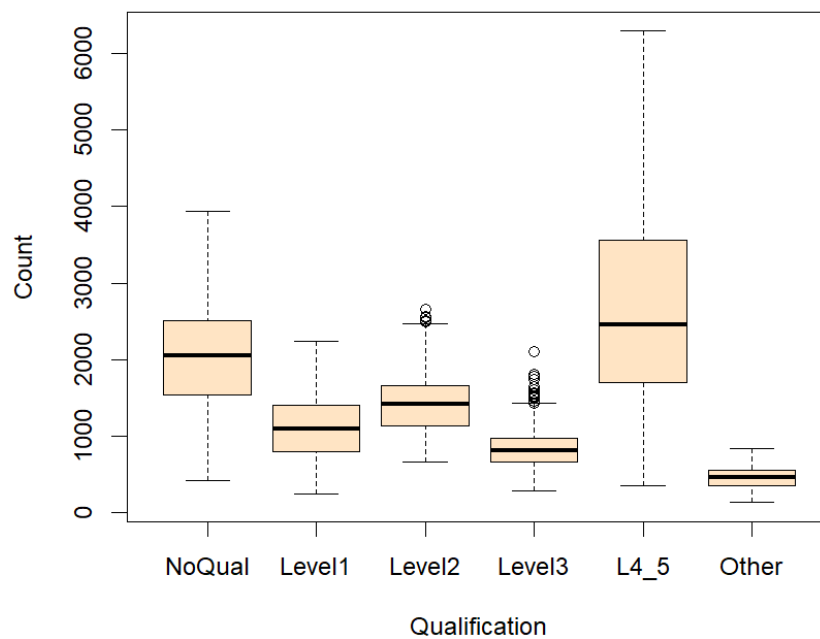
The labeled boxplot identified the outliers as:

- Case 20 : Childs Hill (Borough of Barnet): 12,761 people aged above 16.
- Case 92 : Darwin (Borough of Bromley): 3,552 people aged above 16.

The outliers indicate real demographic contrasts within London. Childs Hill is a densely populated urban area, explaining its high value, while Darwin is a more rural and sparsely populated ward. Therefore, the variation shown in the boxplots reflects true population differences across wards rather than anomalies in the dataset.

Section 6

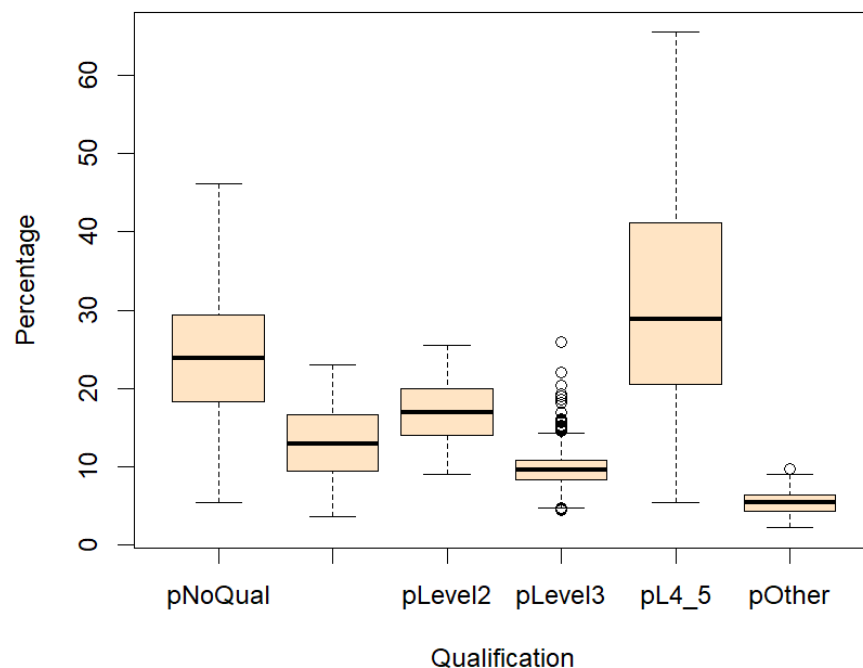
Boxplot of All Six Count Variables



Interpretation

The first boxplot shows the counts of people in each qualification category. Among these, Level 4–5 stands out with the largest values and widest range, meaning some wards have a much higher number of highly qualified residents. No Qualification also shows considerable variation, suggesting uneven educational outcomes across wards. The remaining categories have smaller spreads and lower medians, indicating more consistency in those qualification levels across London.

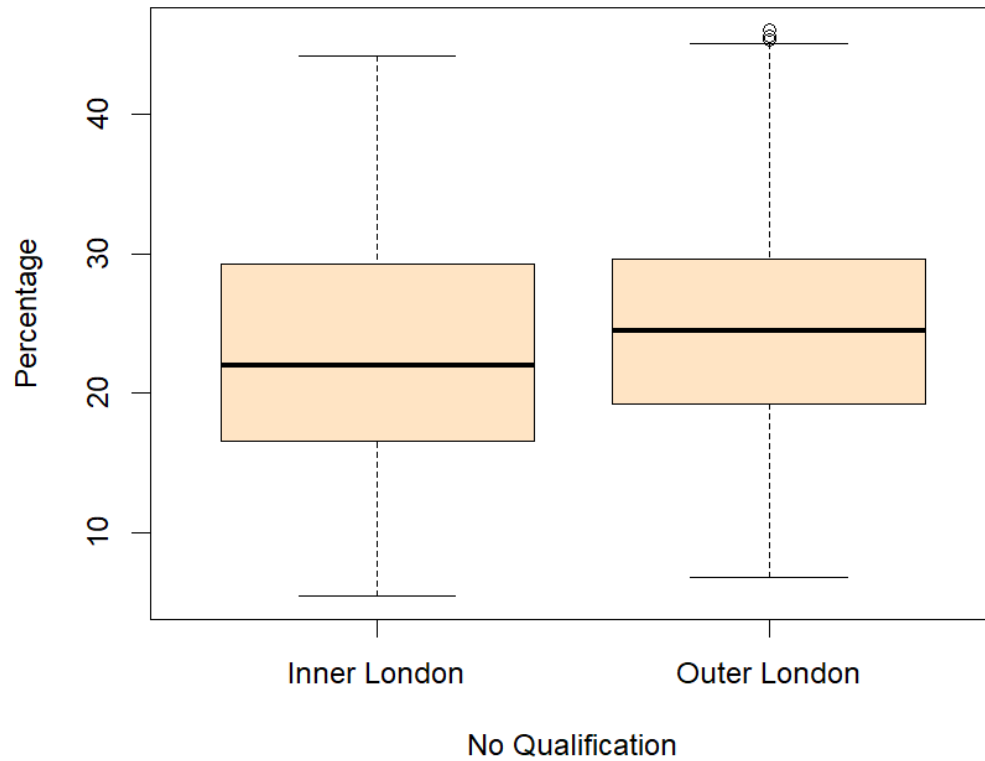
Boxplot of All Six Percentage Variables



Interpretation

When converted to percentages, the pattern remains similar. Level 4–5 again shows the greatest variation, while No Qualification also differs widely between wards. This confirms that the share of residents with higher or no qualifications varies significantly across different parts of London, reflecting its social and educational diversity.

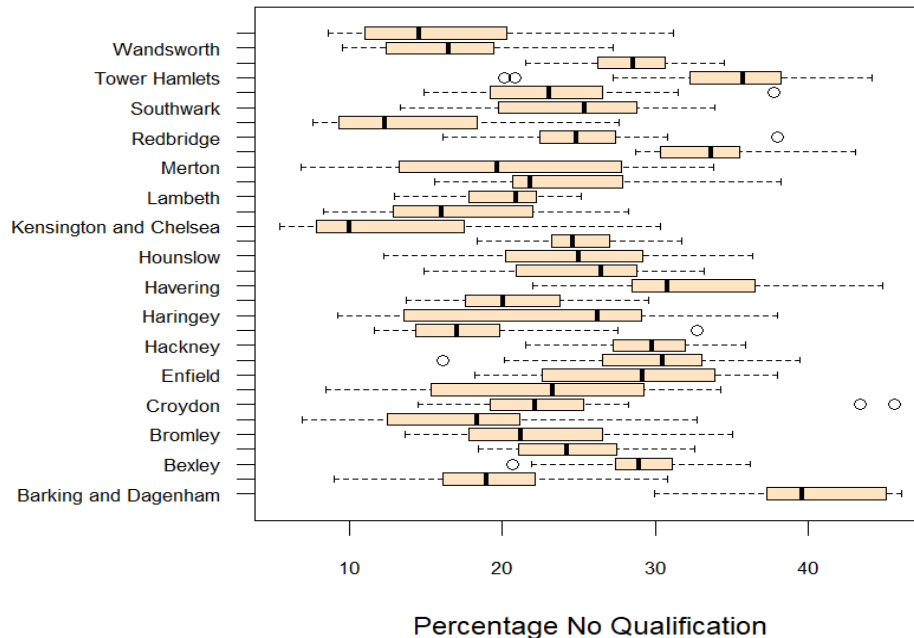
Boxplot of pNoQual by Sub_region



Interpretation

Comparing pNoQual between Inner and Outer London shows slightly higher median percentages of residents without qualifications in Outer London, though both areas overlap considerably. This means educational attainment is broadly comparable, but inner areas tend to have slightly fewer people with no qualifications.

Boxplot of pNoQual by Borough:



Interpretation

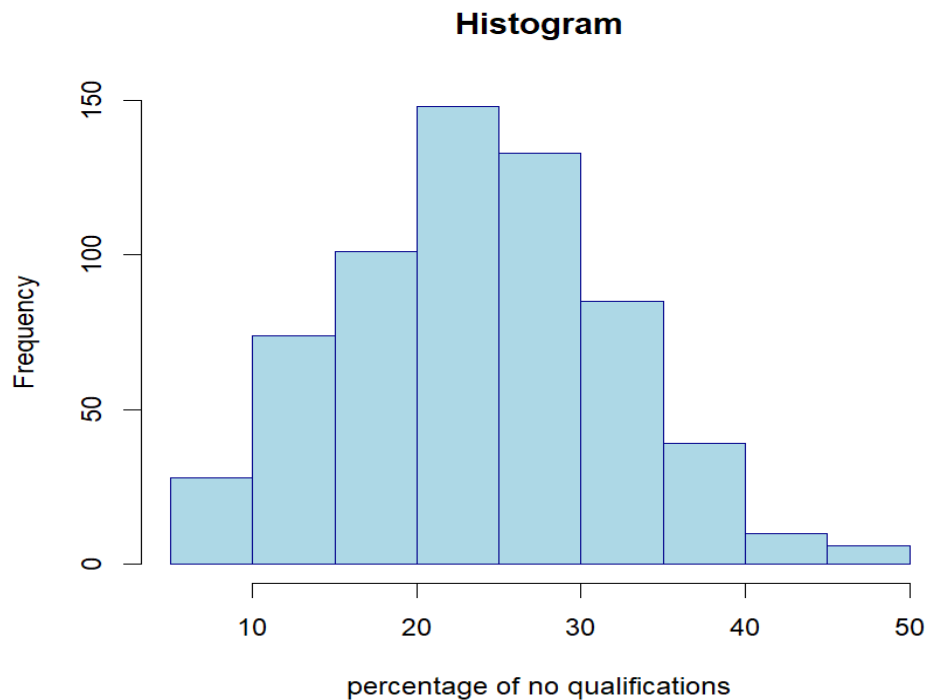
The borough wise plot reveals clear local variation. Boroughs like Barking and Dagenham, Bexley, and Bromley show higher percentages of residents without qualifications, while boroughs such as Wandsworth, Lambeth, and Kensington and Chelsea display lower levels. This suggests that more central and affluent boroughs generally have better qualified populations.

Overall Pattern of pNoQual Across London

Overall, the boxplots indicate a spatial pattern where Inner London wards tend to have higher levels of educational attainment, while certain Outer London boroughs show higher proportions of residents without qualifications. These differences likely reflect broader socioeconomic and demographic contrasts across the city.

Section 7: Create histograms (frequency and probability density) for pNoQual.

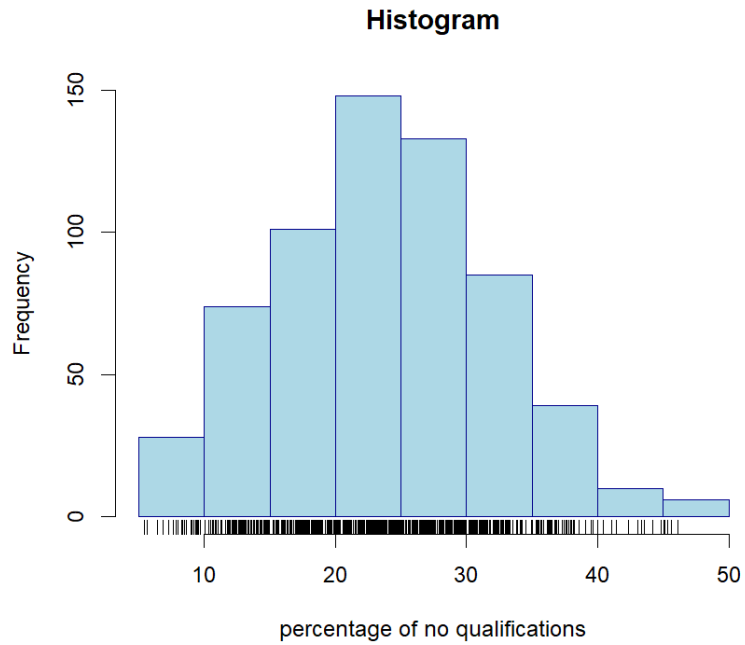
Histogram of the Percentage of Residents with No Qualifications (pNoQual) in London Wards



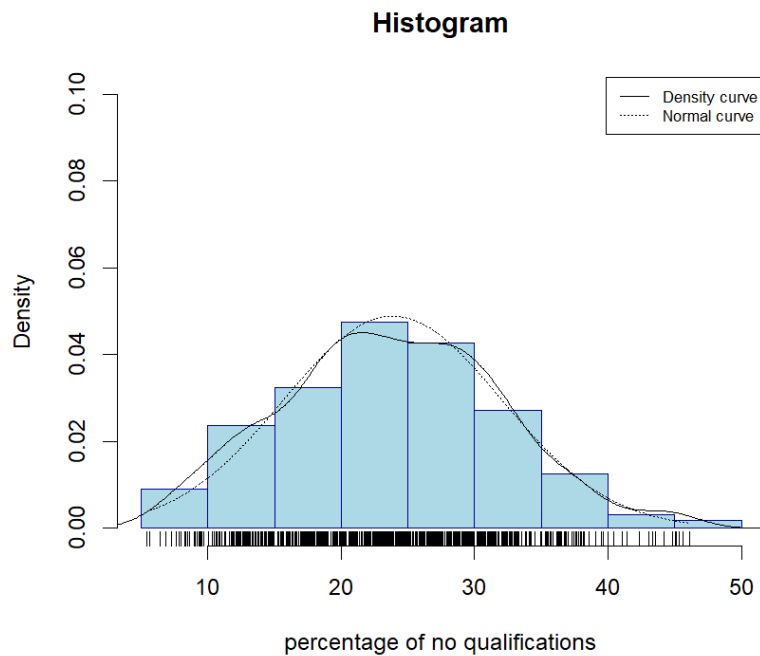
Interpretation:

The frequency histogram shows that most London wards have between 15% and 30% of residents with no qualifications, peaking around 20% to 25%. The distribution is roughly bell-shaped but slightly right-skewed, indicating a few wards with higher percentages. This means that while most wards have similar levels of education, a few areas have higher percentages of people without qualifications.

Histogram with rug



Density Histogram of the Percentage of Residents with No Qualifications (pNoQual)

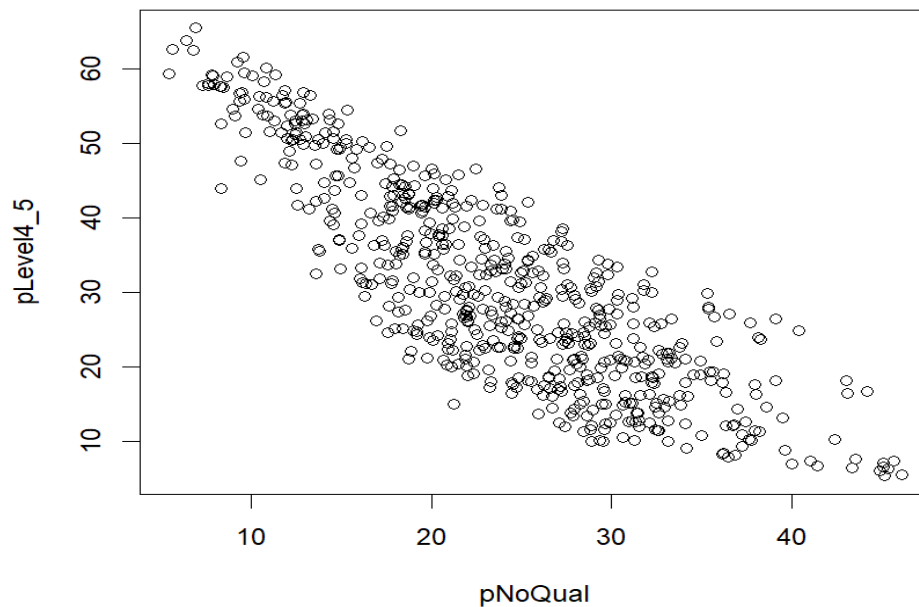


Interpretation

The probability density histogram, with both density and normal curves, shows that pNoQual follows an approximately normal distribution. The curves overlap closely, though a small right skew remains. This means most wards have similar proportions of residents without qualifications, with only a few wards showing notably higher values.

Section 8

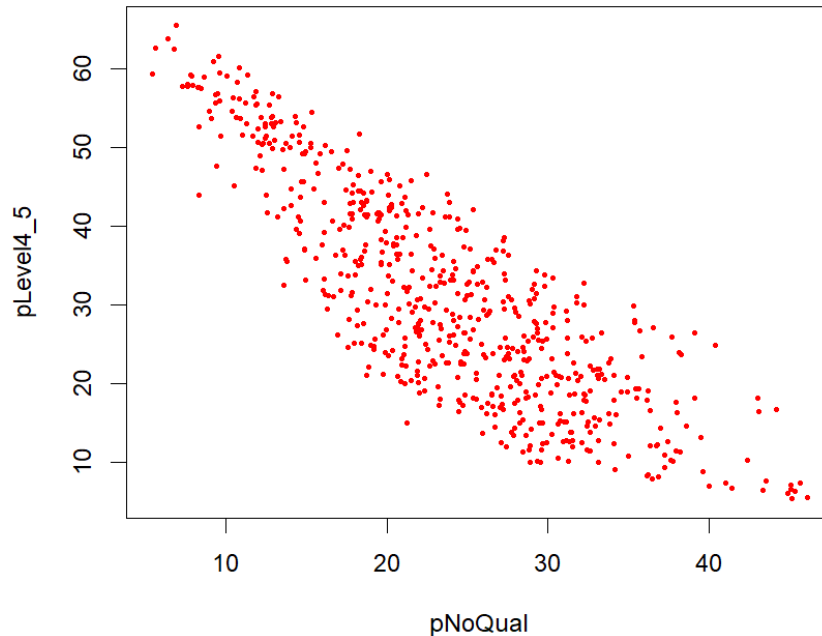
Scatter graph of pNoQual and pLevel4_5



Interpretation

The scatter plot shows a strong negative relationship between pNoQual and pLevel4_5. Wards with a higher percentage of people with no qualifications tend to have a lower percentage of people with Level 4/5 (higher) qualifications, and vice versa.

Second Variant



Interpretation

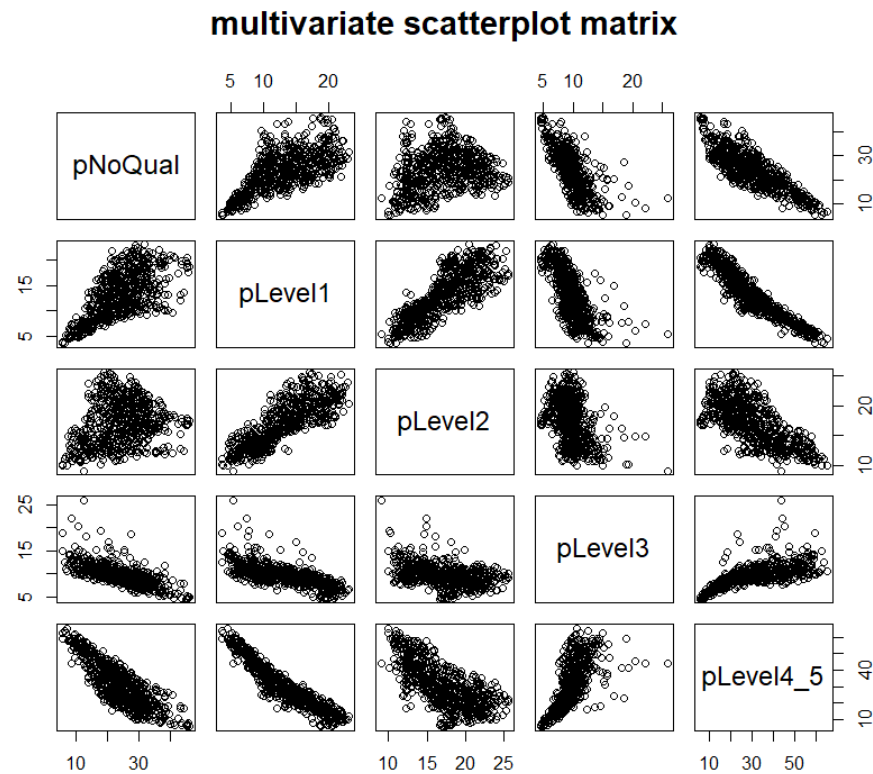
This second scatter plot again shows a strong negative relationship between pNoQual and pLevel4_5. As the percentage of people with no qualifications increases, the percentage with higher-level qualifications clearly decreases. The red points make the downward pattern even easier to see, confirming that wards with more unqualified residents tend to have fewer highly educated residents.

What sort of relationship do these two variables have?

The two variables, pNoQual and pLevel4_5, have a strong negative relationship. This means that as the percentage of people with no qualifications increases, the percentage of people with higher-level qualifications decreases. In other words, wards with more unqualified residents tend to have fewer highly educated individuals.

Section 10

Multivariate Scatterplot matrix



The multivariate scatterplot matrix shows the relationships between all percentage qualification variables which includes pNoQual, pLevel1, pLevel2, pLevel3, and pLevel4_5.

Strongest Positive Correlation

According to the Spearman correlation matrix, the strongest positive correlation is between pLevel1 and pLevel2 ($r = 0.83$), meaning that wards with more residents at Level 1 qualifications usually also have more at Level 2.

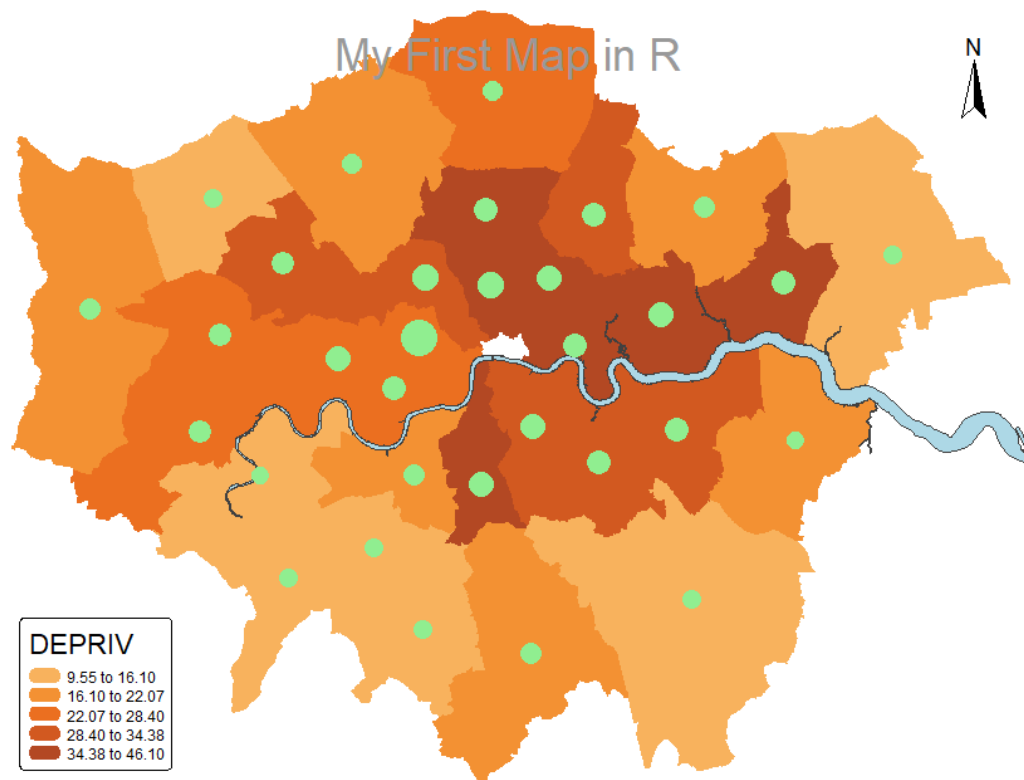
Strongest Negative Correlation

The strongest negative correlation is between pLevel1 and pLevel4_5 ($r = -0.95$), followed closely by pNoQual and pLevel4_5 ($r = -0.83$). This shows that areas with higher numbers of highly qualified residents (Level 4/5) tend to have far fewer people with low or no qualifications.

Overall, the results show a clear educational divide, wards with more highly educated populations tend to have fewer residents with little or no formal education, while lower qualification levels tend to occur together.

Task 2:

Section 4: Plot CRIME_1000 as proportional circles on top of thematic map



Interpretation:

The map shows deprivation levels across London with proportional circles representing crime rates per 1,000 people. The largest circles mostly appear over darker-shaded areas, especially in inner and eastern London, indicating that boroughs with higher deprivation tend to have higher crime rates.

What do you conclude about the spatial relationship of deprivation to crime rate?

This suggests a clear positive spatial relationship between deprivation and crime , areas facing more social disadvantage generally experience more crime incidents.

Reflection

In this session, I learned how to use R to explore and visualize data spatially. Different types of plots, such as boxplots, histograms, scatterplots, and maps were created to understand relationships and patterns within the data. I also learned how to use correlation matrices to check the strength and direction of relationships between variables.

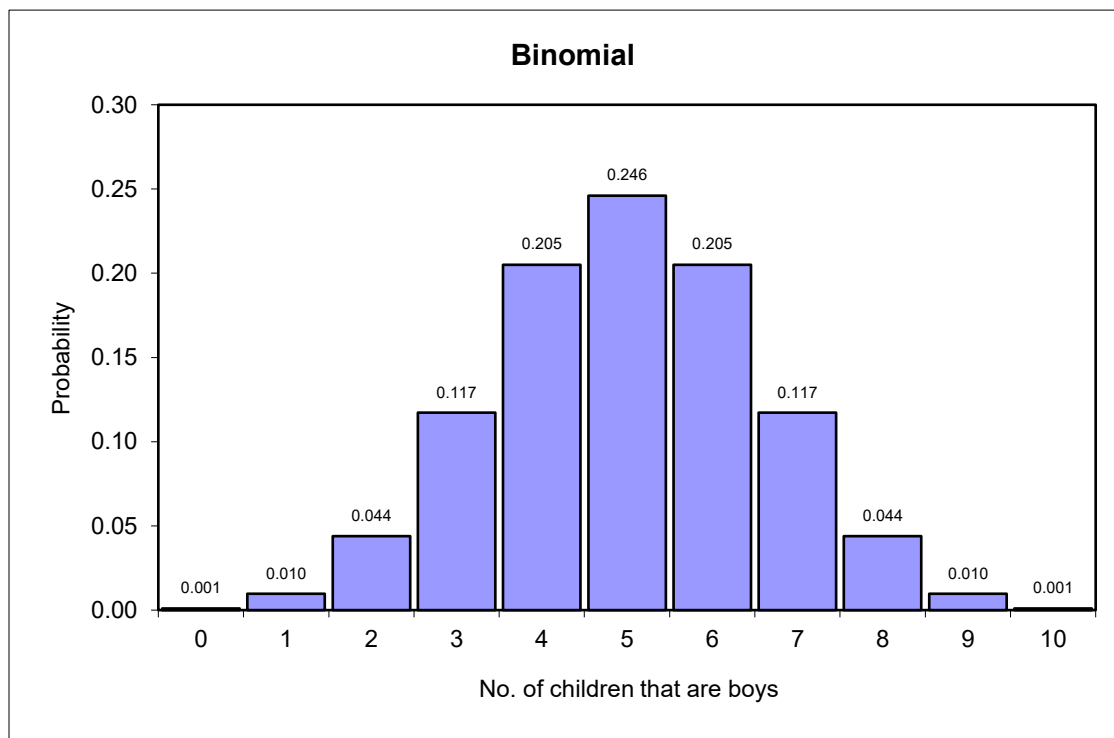
The mapping part was especially useful, as I learned how to create thematic maps and add proportional circles to show multiple layers of information, such as deprivation and crime rates.

Session 5

Probability distributions

Task 1: Binomial Distribution

Graph



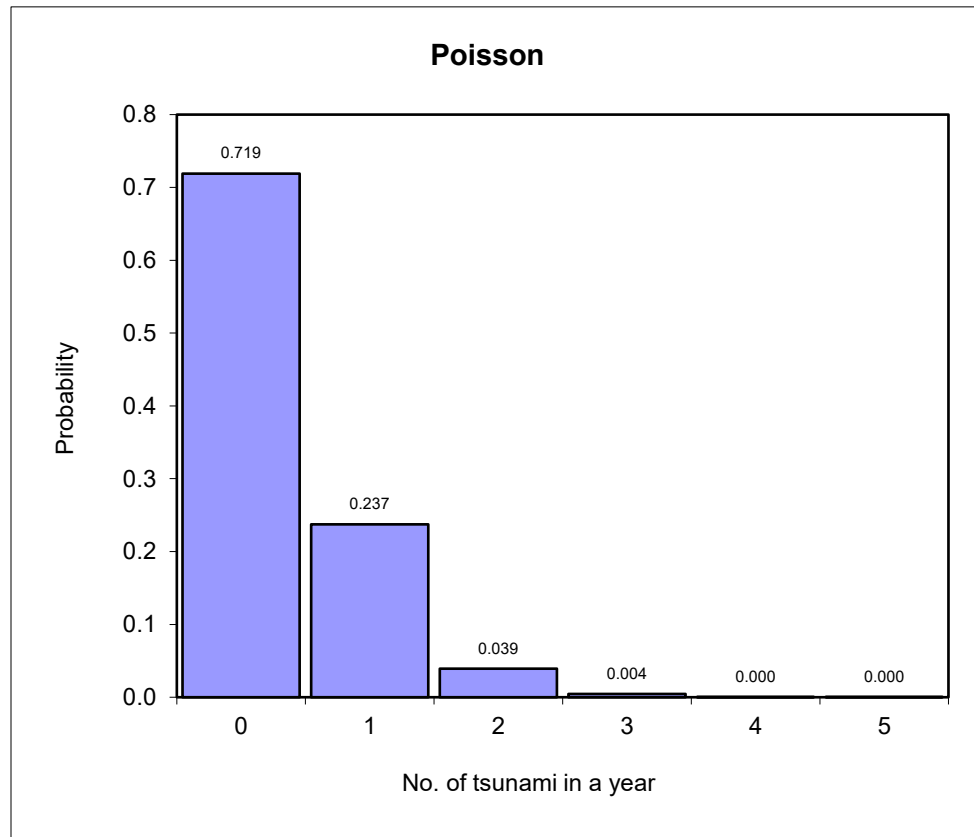
Answer

Using the Binomial distribution, the probability that Jack and Jill have 7 boys out of 10 children (with each child having a 0.5 chance of being a boy) was calculated as 0.117 or 11.7% using `=BINOM.DIST(7,10,0.5,FALSE)`.

The most likely outcome is 5 boys, since it has the highest probability (0.246) and equals the expected value $n \times p = 5$.

Task 2 : Poisson Distribution

Graph:



Answer:

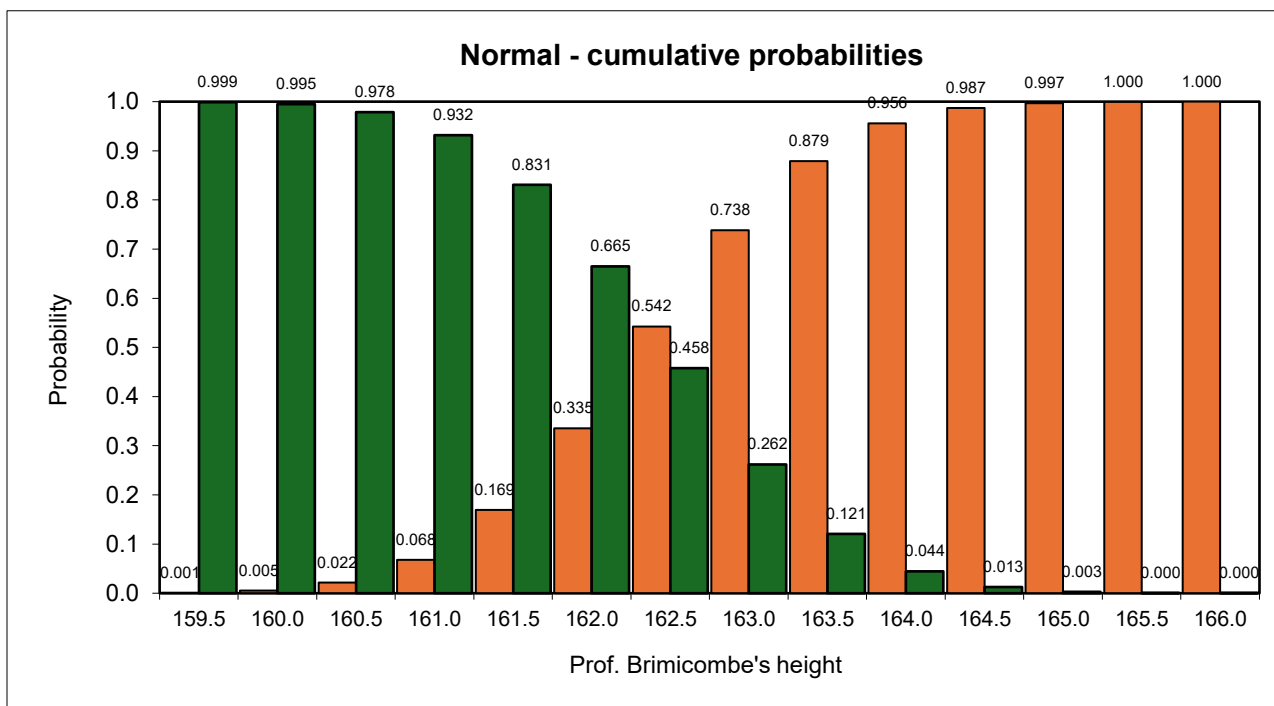
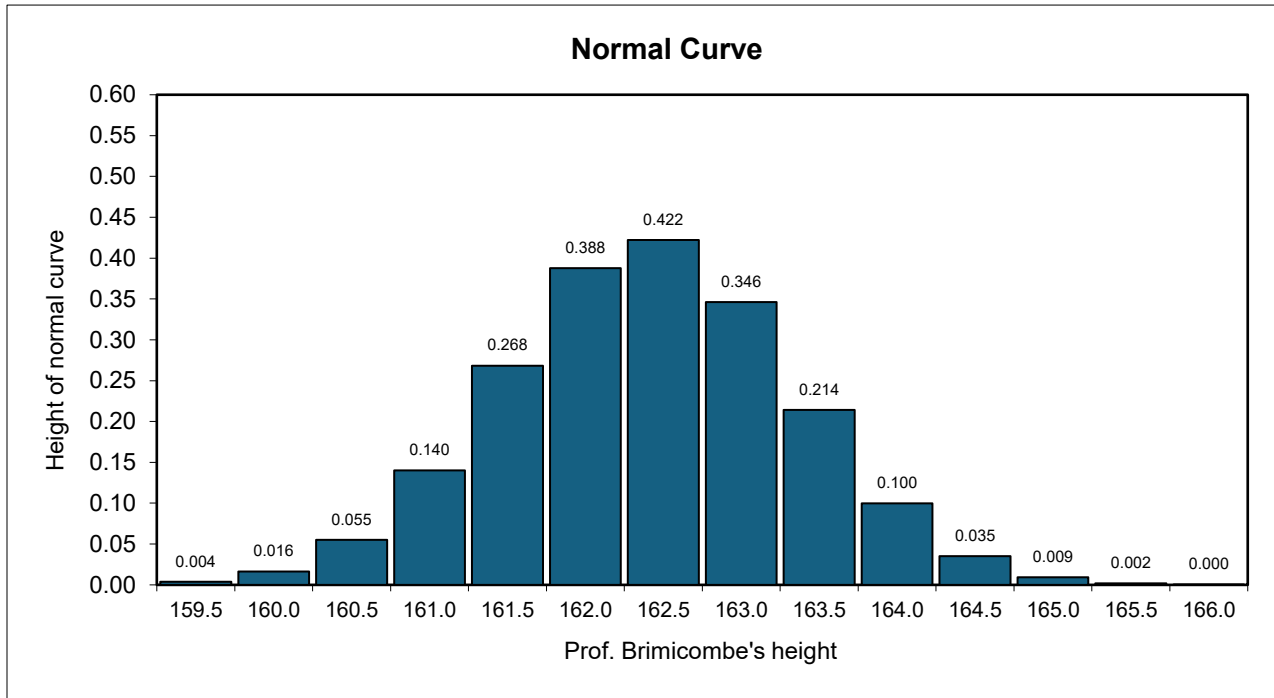
The Poisson distribution was used to find the probability of one tsunami in a year, given an average rate of one every three years ($\lambda = 1/3$).

Using `=POISSON.DIST(A3,0.33,FALSE)`, the probability is 0.239 or 23.9%.

This shows that while tsunamis are rare, there is still a moderate chance of one occurring in a given year.

Task 3: Normal Distribution

Graphs



Answer

Professor Brimicombe's height is normally distributed with a mean of 162.4 cm and a standard deviation of 0.94 cm.

Using $=1 - \text{NORM.DIST}(G3, 162.4, 0.94, \text{TRUE})$, the probability that he is taller than 164 cm is 0.044 or 4.4%.

This means it is quite uncommon for his height to exceed 164 cm.

Reflection

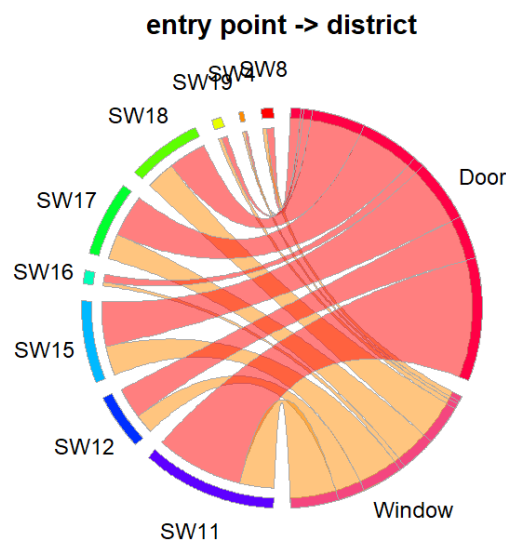
Through this session, I learned how to apply three key probability distributions in Excel. These probability distributions include binomial for discrete outcomes like gender in multiple births, poisson for rare events like tsunamis, and normal for continuous variables like human height.

Each function provides a practical way to model uncertainty and calculate real world probabilities, reinforcing how statistical methods can describe random events accurately.

Session 6

Hypothesis testing (non-parametric)

Circular Plot Showing Relationship Between Entry Point and District (Burglary Data)



Lenovo/2025-10-22

Interpretation

This plot shows how burglars entered homes (through the door or window) in different districts. Most lines connect to Door, meaning doors were the main entry point in most areas, especially in districts like SW11, SW15, and SW17. Window entries happened less often.

Overall, the plot suggests that the method of entry varies between districts, but breaking in through doors was generally more common.

Pearson's Chi-squared test

Result

Pearson's Chi-squared test

data: mytable

X-squared = 54.193, df = 8, p-value = 6.333e-09

Null hypothesis (H₀):

There is no association between the method of entry (Door or Window) and the district.

Explanation

Since the p-value is much smaller than 0.05, we **reject** the null hypothesis.

There is a significant association between entry point and district. This means the way burglars entered (through doors or windows) varied across different districts. Some districts had more door entries, while others had more window entries, as shown in the circular plot.

Chi-squared test using Monte Carlo simulation:

Result

Pearson's Chi-squared test with simulated p-value (based on 2000 replicates)

data: mytable

X-squared = 54.193, df = NA, p-value = 0.0004998

Null hypothesis (H₀):

There is no association between the method of entry (Door or Window) and the district.

Explanation

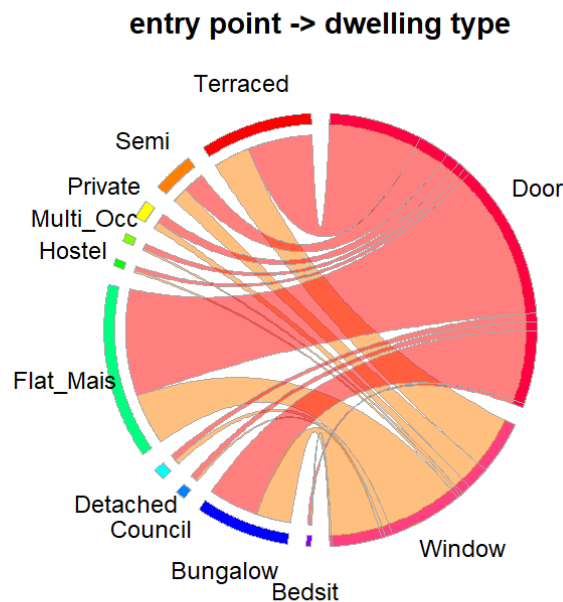
Since the p-value is less than 0.05, we **reject** the null hypothesis.

The result confirms a significant association between the type of entry and the district. The method of entry used by burglars differs by area, suggesting that certain districts experienced more door or window entries than others.

Effect Size (Cramer's $V = 0.118$)

This indicates a small effect size, meaning that although the relationship is statistically significant, the strength of association between entry point and district is relatively weak.

Plot 2: Circular Plot Showing Relationship Between Entry Point and Dwelling Type



Lenovo/2025-10-22

Interpretation

This plot shows how burglars entered different types of dwellings. Most burglaries happened through doors, especially in terraced houses and Flat_Mais. Window entries were less common but still occurred across several dwelling types.

Overall, the plot suggests that the method of entry varies by housing type, with doors being the main entry point in nearly all types of dwellings.

Pearson's Chi-squared Test: Relationship Between Entry Point and Dwelling Type

Result:

Pearson's Chi-squared test

data: mytable

X-squared = 36.866, df = 9, p-value = 2.78e-05

Null Hypothesis (H_0):

There is no association between the type of entry (Door or Window) and the dwelling type.

Explanation:

Since the p-value is well below 0.05, we **reject** the null hypothesis.

There is a significant association between entry method and dwelling type. This means burglars preferred entry points differ by housing category.

For example, Flat_Mais and bungalows show strong deviations from expected values, suggesting that doors and windows are not used equally across all dwelling types.

Chi-Squared Test (Monte Carlo Simulation): Relationship Between Entry Point and Dwelling Type

Result:

Pearson's Chi-squared test with simulated p-value (based on 2000 replicates)

data: mytable

X-squared = 36.866, df = NA, p-value = 0.0004998

Null Hypothesis (H_0):

There is no association between the method of entry (Door or Window) and dwelling type

Explanation

Since the p-value is less than 0.05, we **reject** the null hypothesis.

The result indicates a statistically significant association between entry method and dwelling type. This suggests that burglars' choice of entry point differs depending on the kind of dwelling (e.g., flats, bungalows, or terraced houses).

Effect Size (Cramer's $V = 0.098$):

The effect size is small, meaning that although the association is significant, the strength of the relationship between entry point and dwelling type is weak.

Wilcoxon Signed Rank Test: Male and Female A&E Attendance

Result:

```
wilcoxon signed rank test with continuity correction
```

```
data: AandE$male and AandE$female
```

```
V = 81.5, p-value = 0.0003899
```

```
alternative hypothesis: true location shift is not equal to 0
```

Null Hypothesis (H_0):

There is no difference between male and female A&E attendance figures.

Explanation

Since the p-value is less than 0.05, we reject the null hypothesis.

There is a significant difference between male and female A&E attendances. The summary statistics show that female attendances are generally higher than male attendances across the dataset.

Effect Size ($r = 0.62$):

This represents a medium effect size, indicating a noticeable difference in attendance patterns between males and females.

Mann–Whitney U Test(Wilcoxon rank sum test): Total A&E Attendance by Inner and Outer London

Result

```
wilcoxon rank sum exact test  
  
data: AandE$total by AandE$in_out  
W = 139, p-value = 0.8434  
alternative hypothesis: true location shift is not equal to 0
```

Null Hypothesis (H_0):

There is no difference between male and female A&E attendance figures.

Explanation

Since the p-value is greater than 0.05, we fail to reject the null hypothesis.

There is no statistically significant difference in total A&E attendance between Inner and Outer London.

The summary statistics show that both areas have similar median and mean values, suggesting that overall attendance levels are comparable across the two regions.

Wilcoxon Signed Rank Test: Proportional Male and Female A&E Attendance

Result:

```
wilcoxon signed rank exact test  
  
data: AandE$pMale and AandE$pFemale  
V = 340, p-value = 0.296  
alternative hypothesis: true location shift is not equal to 0
```

Null Hypothesis (H_0):

There is no difference between proportional male and female A&E attendance rates per thousand population.

Explanation:

Since the p-value is greater than 0.05, we fail to reject the null hypothesis.

There is no significant difference between male and female A&E attendance rates once population size is considered.

This suggests that, proportionally, males and females attend A&E at similar rates across London.

Mann–Whitney U Test: Proportional Total A&E Attendance by Inner and Outer London**Result:**

```
wilcoxon rank sum exact test
```

```
data: AandE$pTotal by AandE$in_out
```

```
W = 203, p-value = 0.009935
```

```
alternative hypothesis: true location shift is not equal to 0
```

Null Hypothesis (H_0):

There is no difference in proportional total A&E attendance rates between Inner and Outer London.

Explanation:

Since the p-value is less than 0.05, we reject the null hypothesis.

There is a significant difference in proportional A&E attendance between Inner and Outer London.

The summary statistics show that Inner London has higher proportional attendance rates compared to Outer London.

Effect Size ($r = 0.45$):

This represents a medium effect, meaning the difference between the two regions is noticeable but not large.

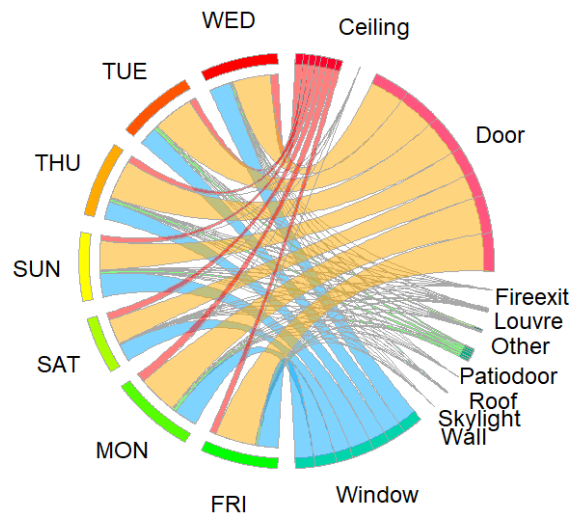
Taking a proportion instead of a count

When using raw counts, the tests showed no significant difference in total A&E attendance between Inner and Outer London. However, when the data were converted to proportions (per thousand population), a significant difference appeared.

This shows that areas with larger populations naturally have higher attendance counts, which can hide real differences in usage rates. By using proportional data, the analysis accounts for population size, giving a fairer comparison. It reveals that, relative to their populations, Inner London has higher A&E attendance rates than Outer London.

Relationship Between Entry Point and Day of the Week

Relationship Between Entry Point and Day of the Week



Lenovo/2025-11-05

Plot Interpretation

The circular plot shows that door and window entries dominate across all days, with only minor variations between weekdays and weekends.

Chi-Squared Test: Relationship Between Entry Point and Day of the Week

Result:

```
Pearson's Chi-squared test
```

```
data: myxtab
```

```
X-squared = 113.24, df = 60, p-value = 3.951e-05
```

```
> Cramerv(myxtab, conf.level = 0.95)
```

```
  Cramer V      lwr.ci      upr.ci  
0.06373045 0.02846539 0.05814234
```

Null Hypothesis (H_0):

There is no association between the type of entry point and the day of the week.

Interpretation

Since the p-value is less than 0.05, we reject the null hypothesis. There is a significant association between entry point and day of the week, meaning that the type of entry used by burglars varies slightly across days.

However, the effect size (Cramer's $V = 0.064$) is very small, indicating that the relationship is weak and that the differences are not practically large.

Session 7

Hypothesis testing (parametric)

Section 02

Descriptive Statistics

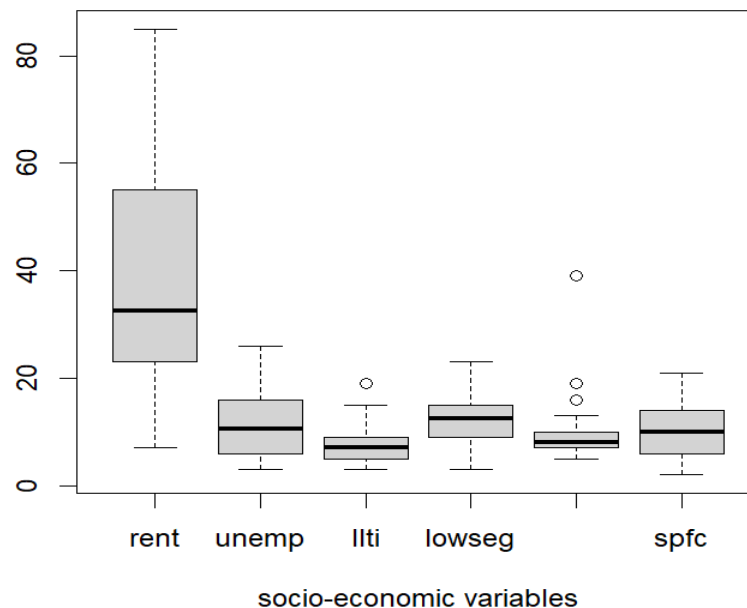
Variable	Minimum	1st Quartile	Median	Mean	3rd Quartile	Maximum
rent	7.00	23.25	32.50	35.94	53.50	85.00
unemp	3.00	6.25	10.50	11.58	16.00	26.00
liti	3.00	5.00	7.00	7.74	9.00	19.00
lowseg	3.00	9.25	12.50	12.44	15.00	23.00
mobile	5.00	7.00	8.00	9.34	9.75	39.00
spfc	2.00	6.00	10.00	10.70	14.00	21.00

Interpretation

The descriptive statistics show that rent, liti, and mobile have mean and median values that are quite close, indicating approximate normality.

In contrast, unemp, lowseg, and spfc show larger differences between mean and median values and wider ranges, suggesting possible skewness and non-normal distributions.

Boxplot



Interpretation

The boxplot compares the distribution of six socio-economic variables which include rent, unemployment (unemp), limiting long-term illness (llti), low-segment employment (lowseg), mobility (mobile), and single-parent families with children (spfc).

The variable rent shows a wide range but appears relatively balanced around the median, suggesting it is close to a normal distribution.

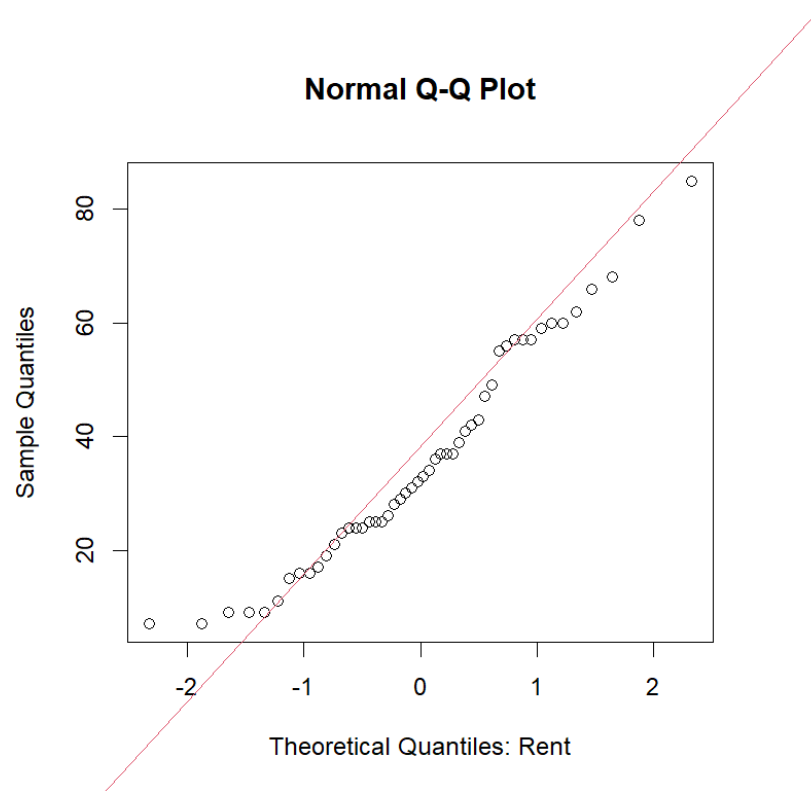
In contrast, unemp, lowseg, and spfc have longer upper whiskers and several outliers, which indicate positive skewness and a departure from normality.

The llti and mobile variables look more symmetric and compact, suggesting that their data are distributed more evenly.

Overall, the boxplot suggests that unemp, lowseg, and spfc are likely non-normal, while rent, llti, and mobile appear approximately normal.

Section 4

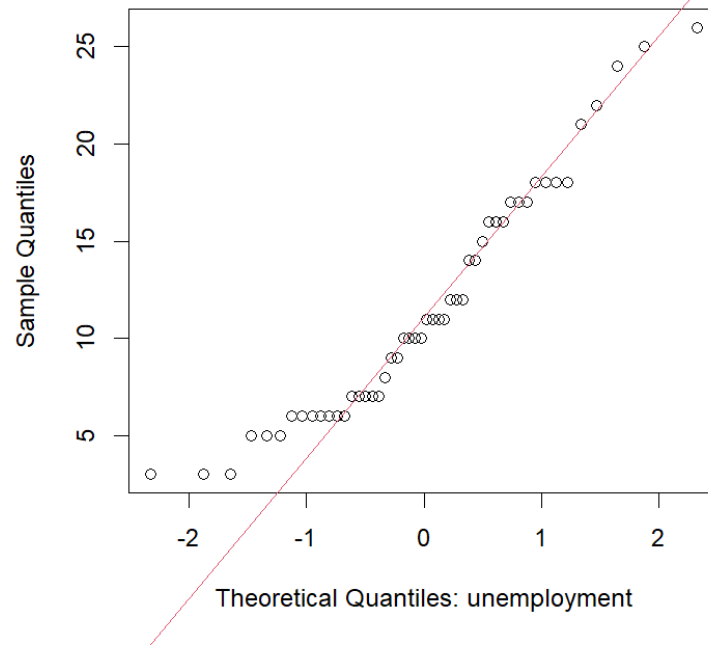
Output



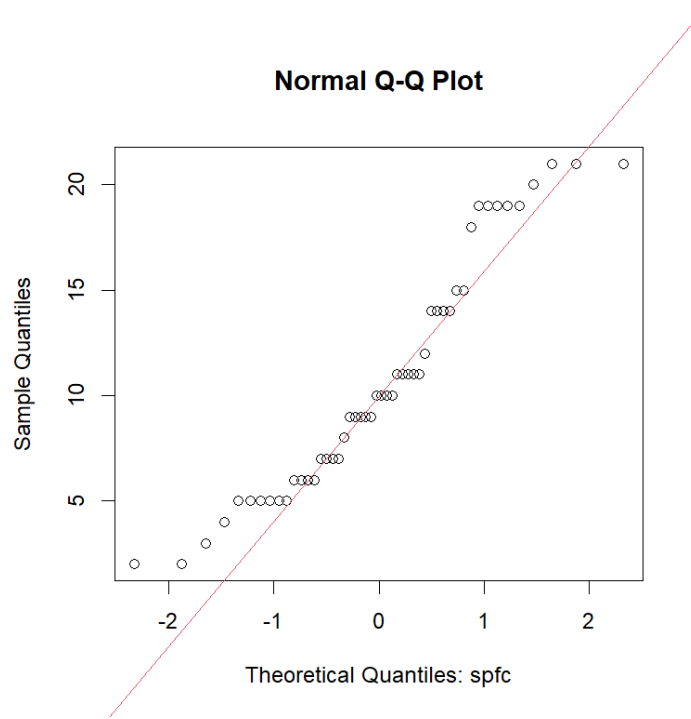
Interpretation

The Q-Q plots display how closely the sample data follow a normal distribution. The plot for rent shows most of the points lying close to the red diagonal line, indicating that the data are approximately normally distributed with only slight deviations at the extremes.

Normal Q-Q Plot



Normal Q-Q Plot



Interpretation

In contrast, the plots for unemployment (unemp) and single-parent families with children (spfc) show more noticeable departures from the straight line, particularly at the upper end, where points rise above the line. This pattern suggests positive skewness and a non-normal distribution for these two variables.

Overall, these plots confirm that rent follows a roughly normal pattern, while unemp and spfc deviate from normality, supporting the conclusions drawn from the descriptive statistics and boxplots.

Section 05 : Normality Tests

Results of Kolmogorov–Smirnov and Shapiro–Wilk Tests for Normality

Variable	Test	Test Statistic	p-value	Interpretation
rent	KS	D = 0.098	0.7178	Normal
unemp	KS	D = 0.136	0.3138	Normal
llti	KS	D = 0.182	0.073	Normal (borderline)
lowseg	KS	D = 0.061	0.9928	Normal
mobile	KS	D = 0.267	0.0016	Non-normal
spfc	KS	D = 0.138	0.2937	Normal
rent	Shapiro–Wilk	W = 0.955	0.056	Normal
llti	Shapiro–Wilk	W = 0.887	0.00018	Non-normal
mobile	Shapiro–Wilk	W = 0.533	2.2e-11	Non-normal

Interpretation

The results of the Kolmogorov–Smirnov and Shapiro–Wilk tests indicate that most variables (rent, unemp, lowseg, and spfc) have p-values above 0.05, suggesting they are approximately normally distributed. However, llti and mobile show very low p-values ($p < 0.05$), meaning they significantly deviate from normality.

These findings confirm that llti and mobile are non-normal, while the other variables can be treated as normally distributed.

Section 6: Paired Samples t-tests

Results of Paired Samples t-tests

Paired t-test

```
data: rent and lowseg
t = 9.7344, df = 49, p-value = 4.882e-13
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 18.64865 28.35135
sample estimates:
mean difference
      23.5
```

Interpretation

The first paired samples t-test compared the mean percentage of households in rented accommodation (rent) and those in low-segment employment (lowseg). The results show a highly significant difference ($t = 9.73$, $p < 0.001$), with rent having a much higher mean value. The effect size (Cohen's $d = 1.65$) indicates a large effect, suggesting a strong difference between these two socio-economic factors.

In contrast, the second paired t-test comparing unemployment (unemp) and single-parent families with children (spfc) shows no significant difference ($t = 1.66$, $p = 0.10$). The mean difference of 0.88 is small, and the confidence interval includes zero, indicating that the two variables are statistically similar in this sample.

Overall, rent and lowseg differ significantly, while unemp and spfc do not.

Section 07 – Wilcoxon Signed Rank Test

Results of Wilcoxon Signed Rank Tests

```
wilcoxon signed rank test with continuity correction
```

```
data: unemp and spfc
```

```
V = 648, p-value = 0.1404
```

```
alternative hypothesis: true location shift is not equal to 0
```

Interpretation:

The Wilcoxon Signed Rank Test, a non-parametric alternative to the paired t-test, was used to verify the earlier results without assuming normality.

The test for rent and lowseg produced a very small p-value ($p < 0.001$), indicating a statistically significant difference between the two variables. This result is consistent with the paired t-test from Section 6, confirming that rent values are significantly higher than lowseg.

For unemp and spfc, the test returned a p-value of 0.14, which is greater than 0.05. This means there is no significant difference between the two variables, supporting the earlier t-test result.

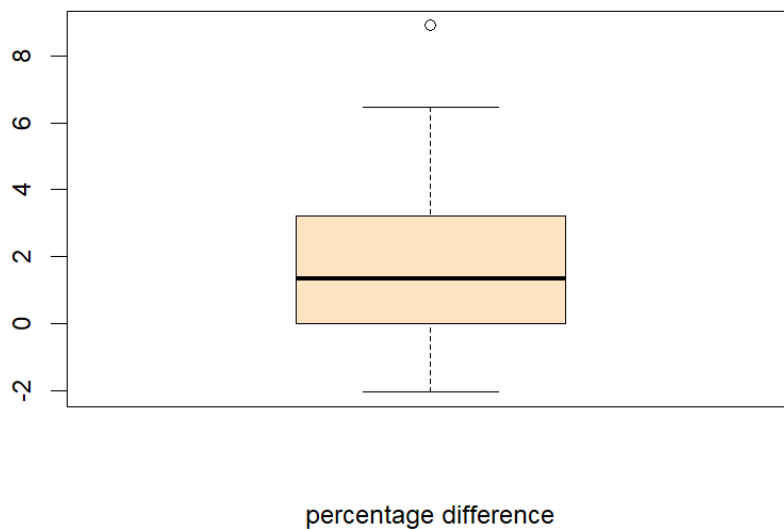
Overall, both parametric and non-parametric tests lead to the same conclusion: rent and lowseg differ significantly, while unemp and spfc do not.

Session 8

t-test and Analysis of Variance

Section 05 :Boxplot and Q-Q Plot of Percentage Difference

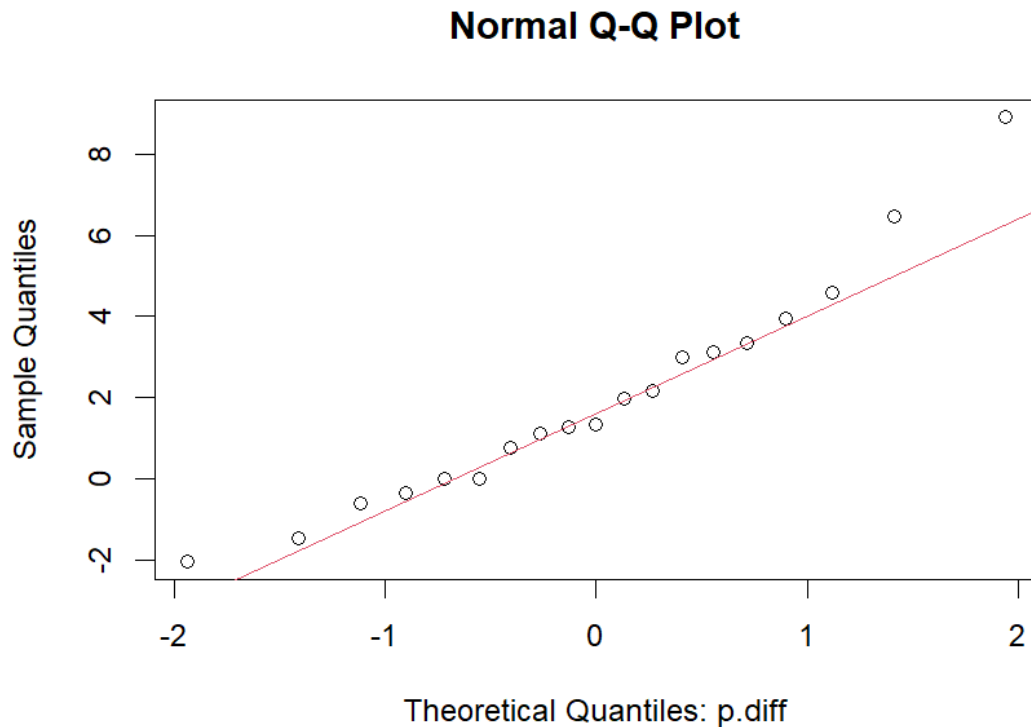
Boxplot:



Interpretation:

The boxplot shows that the data are approximately symmetric around the median, with one mild outlier above 8%. Most observations fall within a narrow range, suggesting the absence of extreme variability.

Q-Q plot:



Interpretation:

The Q-Q plot shows that most points lie close to the red reference line, indicating approximate normality with only minor deviations in the upper tail.

Kolmogorov-Smirnov (K-S) Test for Normality

Output

```
Asymptotic one-sample Kolmogorov-Smirnov test
```

```
data: blood.tests$p.diff  
D = 0.11711, p-value = 0.9568  
alternative hypothesis: two-sided
```

Interpretation

The K–S test returned $p = 0.9568$, which is far greater than 0.05.

Therefore, we fail to reject the null hypothesis that the data are normally distributed.

This confirms that the p.diff variable is suitable for parametric analysis.

One-Sample t-Test on p.diff

Results

```
One sample t-test
```

```
data: blood.tests$p.diff
```

```
t = 3.1637, df = 18, p-value = 0.005374
```

```
alternative hypothesis: true mean is not equal to 0
```

```
95 percent confidence interval:
```

```
0.6646709 3.2925655
```

```
sample estimates:
```

```
mean of x
```

```
1.978618
```

Interpretation

Null hypothesis (H_0): The mean percentage difference = 0

Alternate hypothesis (H_1): The mean percentage difference is not equal to 0

Since $p = 0.005 < 0.05$, we **reject H_0** .

The mean percentage difference of approximately 1.98 % is significantly greater than zero, indicating that freezing has a measurable effect on blood test results.

The effect size (Cohen's $d = 0.73$) represents a moderate-to-large effect.

Section 9: ANOVA test

One-Way ANOVA (by Council)

Results

```
summary(fit1)
      Df Sum Sq Mean Sq F value    Pr(>F)    
council    2   2837   1418.5    20.83 3.34e-07 ***
Residuals  47   3201     68.1                
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

>
```

Interpretation

The one-way ANOVA indicates a highly significant difference in mean educational expenditure across councils

H₀: The mean educational expenditure is the same across all councils.

H₁: At least one council differs in mean expenditure.

Since $p = 3.34 \times 10^{-7} < 0.05$, we **reject H₀**.

This result suggests that political control of the council has a statistically significant effect on educational spending levels.

Tukey HSD (by Council)

Results

```
TukeyHSD(fit1)
Tukey multiple comparisons of means
 95% family-wise confidence level

Fit: aov(formula = ed_exp ~ council)

$council
      diff      lwr      upr    p adj
Lab-Con 18.544118 11.587412 25.500823 0.0000002
Lib-Con  9.470588  2.620098 16.321078 0.0045357
Lib-Lab -9.073529 -16.030235 -2.116824 0.0077054
```

Null Hypothesis (H_0):

There is no difference in mean educational expenditure between the two council groups being compared

Alternative Hypothesis :

There is a difference in mean educational expenditure between the two council groups.

Interpretation

The Tukey HSD test shows statistically significant differences in educational expenditure among all three council types.

Since all p-values < 0.01 , we **reject** H_0 .

Labour led councils have the highest mean expenditure, followed by Liberal, then Conservative, indicating that political affiliation has a significant impact on educational spending levels.

ANOVA for Educational Expenditure by Council and Nation

Results

```
summary(fit2)
      Df Sum Sq Mean Sq F value    Pr(>F)
council    2   2837   1418.5    51.44 1.86e-12 ***
eng_wal    1   1932   1932.4    70.07 8.45e-11 ***
Residuals 46   1269     27.6
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

>
```

Effect of Council on Educational Expenditure

Null Hypothesis (H_{01}):

There is no difference in mean educational expenditure among the three council types.

Alternative Hypothesis (H_{11}):

At least one council type has a different mean expenditure.

Interpretation:

Since $p < 0.05$, the test shows a highly significant difference in educational expenditure among the three council types therefore we **reject** H_{01} and **accept** H_{11} . This means political control of the council has a statistically significant effect on educational spending.

Effect of Nation (eng_wal) on Educational Expenditure

Null Hypothesis (H_{02}):

There is no difference in mean educational expenditure between England and Wales.

Alternative Hypothesis (H_{12}):

There is a difference between the mean expenditures of the two nations.

Explanation:

Since $p < 0.05$, this result is also highly significant, indicating a clear difference in expenditure between the two nations, we **reject** H_{02} and **accept** H_{12} .

This mean educational expenditure differs significantly between England and Wales.

Tukey HSD Test for Educational Expenditure by Council and Nation

Results

```
> TukeyHSD(fit2)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = ed_exp ~ council + eng_wal)

$council
      diff      lwr      upr    p adj
Lab-Con 18.544118 14.114139 22.97410 0.00e+00
Lib-Con  9.470588  5.108246 13.83293 1.09e-05
Lib-Lab -9.073529 -13.503508 -4.64355 2.95e-05

$eng_wal
      diff      lwr      upr    p adj
W-E 12.41882  9.428956 15.40869      0
```

Null Hypothesis (H₀):

There is no difference in mean educational expenditure between the two groups.

Alternative Hypothesis (H₁):

There is a difference in mean expenditure between the two groups.

Interpretation

The Tukey HSD test indicates significant differences in educational expenditure both among councils and between nations.

All pairwise comparisons for both council and nation are statistically significant ($p < 0.001$) so null hypothesis is rejected for all comparisons.

This means educational expenditure differs significantly between all council types, and educational expenditure is higher in Wales than in England.

These findings reinforce the earlier ANOVA results and show that both political leadership and national factors substantially affect educational spending levels.

Kruskal–Wallis Test Results for Educational Expenditure

```
kruskal.test(ed_exp ~ council, data = Johnston.80)
```

```
kruskal-wallis rank sum test
```

```
data: ed_exp by council
```

```
kruskal-wallis chi-squared = 23.687, df = 2, p-value = 7.186e-06
```

```
> kruskal.test(ed_exp ~ eng_wal, data = Johnston.80)
```

```
kruskal-wallis rank sum test
```

```
data: ed_exp by eng_wal
```

```
kruskal-wallis chi-squared = 14.264, df = 1, p-value = 0.0001589
```

Kruskal–Wallis Test: Educational Expenditure by Council

Null Hypothesis (H_0):

The median educational expenditure is the same for all three council groups (Labour, Liberal, Conservative).

Alternative Hypothesis (H_1):

At least one council group has a different median educational expenditure.

Explanation:

Since the p-value is far below 0.05, the results show a significant difference in median educational expenditure across the three council groups.

Therefore, the null hypothesis is rejected.

This confirms that the level of educational spending varies depending on which political party controls the council.

This finding is consistent with the earlier ANOVA and Tukey results.

Session 9

Factor Analysis and Cluster Analysis

A table constructed from Section04 and Section05 that compares the correlations, the partial correlations and their p values. Comment on what you find.

Variables	Control Variable	Spearman's r	p-value	Partial r	p-value
Life_Male – Dom_Build	None	-0.281	0.1189	–	–
Life_Male – Smoking	None	-0.610	0.0002	-0.334	0.0662
Life_Male – Obese	None	-0.200	0.273	–	–
Life_Male – Episodes	None	-0.316	0.0779	–	–
Life_Male – Benefits	None	-0.870	< 0.001	-0.810	< 0.001
Life_Male – Crime	None	-0.611	0.0002	–	–

Interpretation

The correlation analysis shows that male life expectancy is negatively associated with several socioeconomic and health-related factors in London districts. The strongest relationship is observed between life expectancy and the proportion of residents claiming benefits ($r = -0.87$, $p < 0.001$). This indicates that areas with higher benefit dependency tend to have lower male life expectancy, suggesting that socioeconomic deprivation is a significant determinant of health outcomes.

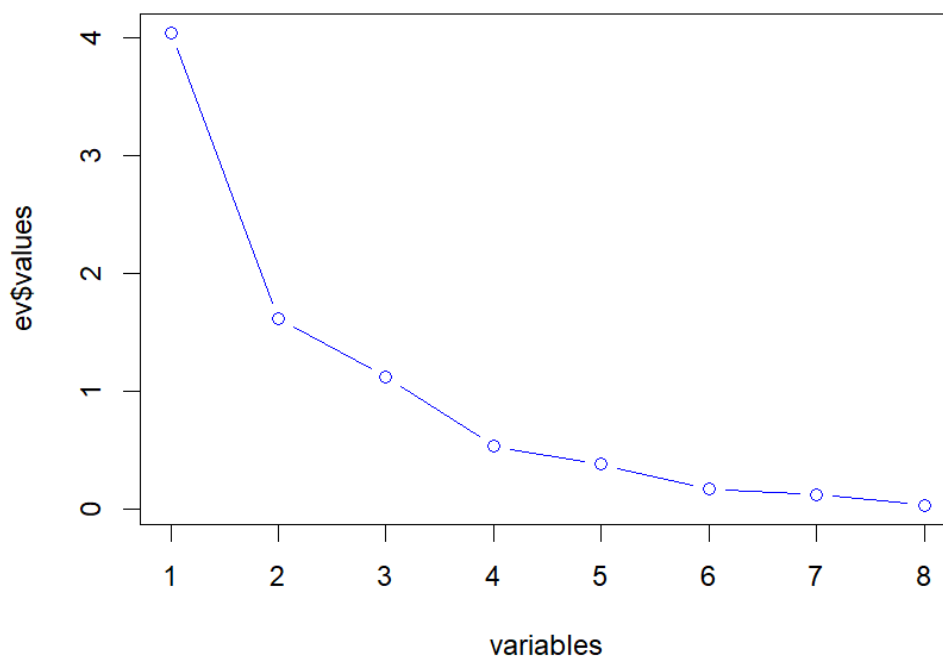
Smoking and crime rates also display significant negative correlations with life expectancy ($r = -0.61$, $p < 0.001$ for both). Obesity, domestic building density, and the number of hospital episodes are weakly correlated and not statistically significant.

The partial correlation results provide additional insight. When controlling for smoking, the relationship between life expectancy and benefits remains strong (partial $r = -0.81$, $p < 0.001$). This suggests that the association between socioeconomic deprivation and life expectancy is largely independent of smoking behavior. However, when controlling for benefits, the correlation between life expectancy and smoking decreases substantially (partial $r = -0.33$, $p = 0.066$), indicating that the influence of smoking is partly explained by underlying socioeconomic conditions.

Overall, the findings suggest that socioeconomic deprivation is the most influential factor associated with male life expectancy disparities across London districts, while smoking appears to act as a mediating variable linked to deprivation rather than an entirely independent predictor.

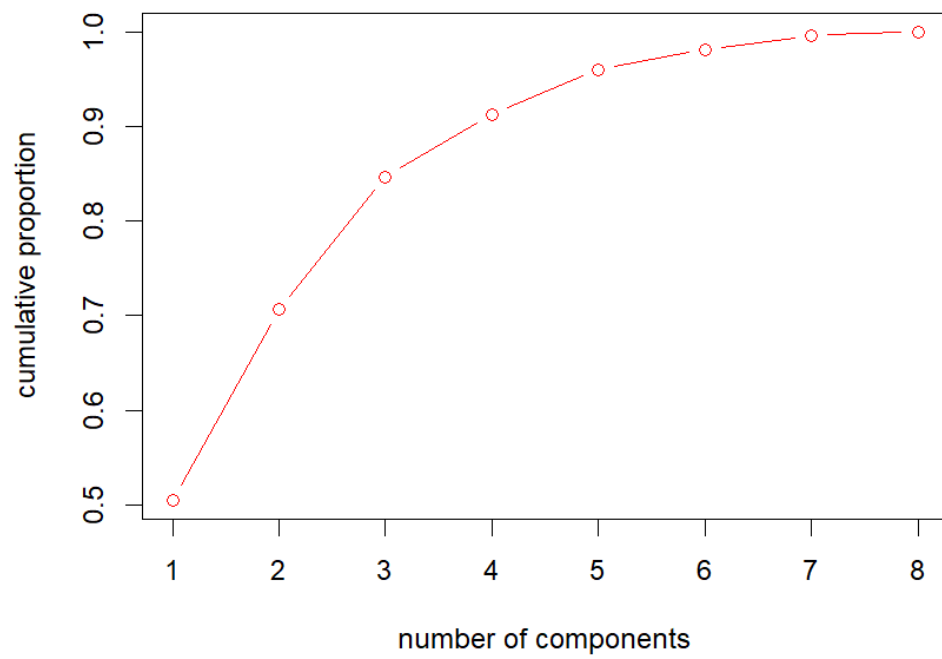
The screeplot and plot showing cumulative percentage

A Principal Components Analysis (PCA) was conducted on eight socioeconomic and health related variables for London districts. The Kaiser–Meyer–Olkin (KMO) measure of 0.73 indicated that the data were suitable for PCA.



Interpretation:

The scree plot shows a clear inflection after the fourth component, suggesting that four components should be retained.



Interpretation

The cumulative variance plot confirms that these four components together explain approximately 91 % of the total variance in the dataset.

Rotated Component Matrix (Varimax Rotation)

Variable	RC1	RC2	RC3	RC4	Complexity
Dom_Build	0.91	-0.03	0.08	-0.27	1.2
NonDom_Build	0.60	0.09	0.71	-0.31	2.4
Dom_Gardens	0.11	-0.28	-0.84	0.35	1.6
Greenspace	-0.90	-0.16	-0.24	0.14	1.3
Smoking	0.13	0.92	0.18	0.21	1.2
Obese	-0.20	0.06	-0.18	0.92	1.2
Episodes	-0.31	0.37	-0.20	0.73	2.1
Crime	0.45	-0.02	0.86	0.00	1.5

Interpretation of the Four Components

Component 1 : Built Environment and Urban Structure

High loadings for Dom_Build (0.91) and NonDom_Build (0.60) indicate this component represents residential and commercial density.

Negative loading for Greenspace (−0.90) suggests an inverse link between dense urban areas and green space availability.

Component 2: Lifestyle

Dominated by Smoking (0.92), this factor reflects lifestyle risk behaviours, particularly smoking prevalence.

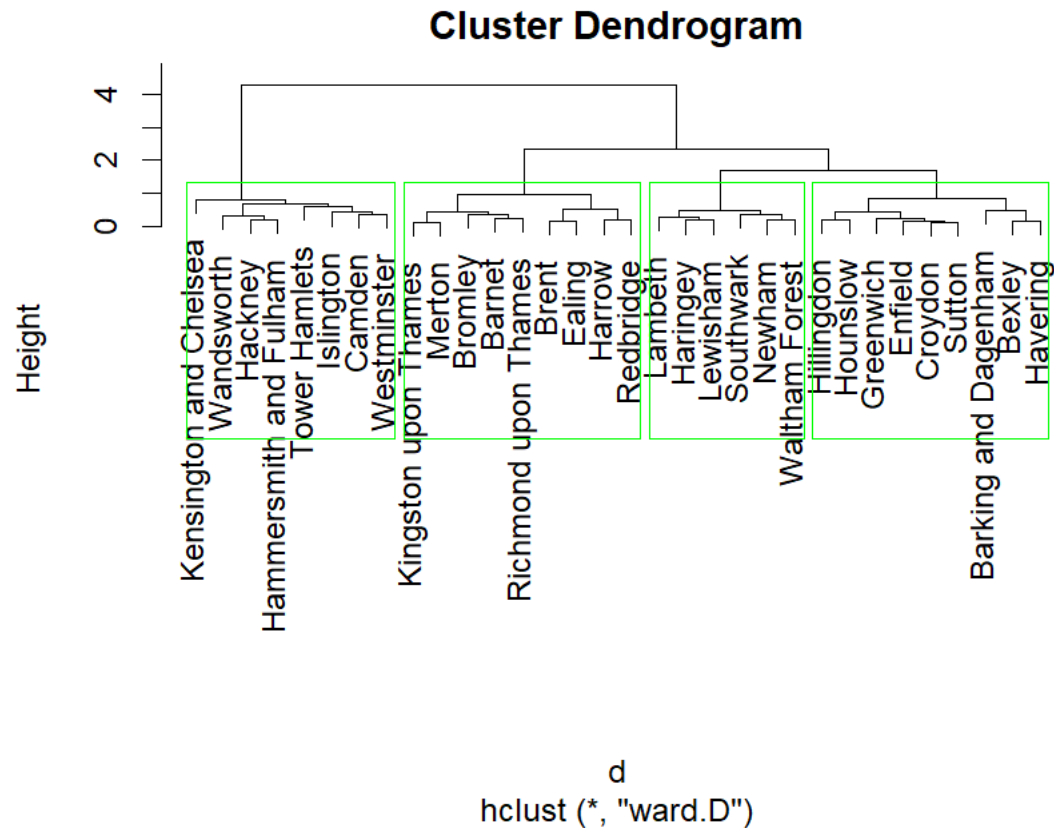
Component 3: Crime and Environmental Stress

Strongly associated with Crime (0.86) and negatively with Dom_Gardens (−0.84), indicating environmental quality and social disorder patterns.

Component 4 : Health Outcomes

High loadings for Obese (0.92) and Episodes (0.73) suggest that this component represents population health conditions linked to obesity and medical service use.

Dendrogram:



Interpretation

The dendrogram shows four main clusters, which were defined by drawing a cut at the height level corresponding to four clusters (green boxes). These clusters represent groups of London boroughs with similar social, health, and environmental profiles:

Cluster 1: Includes highly urbanized inner-London boroughs (for example Kensington and Chelsea, Westminster, Islington) characterized by dense built environments and higher social complexity.

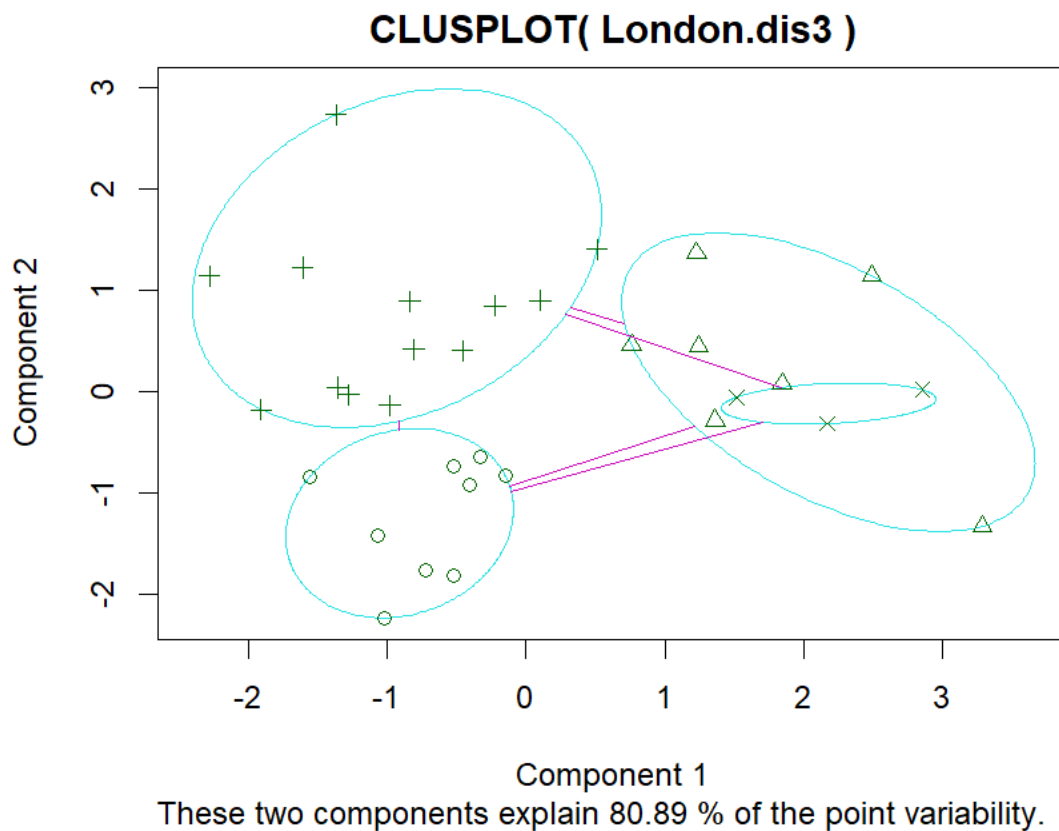
Cluster 2: Suburban boroughs such as Kingston upon Thames and Bromley, showing lower population density and better health indicators.

Cluster 3: Mixed boroughs like Lewisham, Southwark, and Newham, representing moderately urbanized areas with mid-level health risks.

Cluster 4: Outer boroughs such as Croydon, Bexley, and Havering, reflecting more residential, less dense, and healthier environments overall.

This clustering provides a spatial understanding of how environmental and health-related variables group London districts into meaningful categories based on their socioeconomic and health characteristics.

Cluster Visualization (K-Means Method)



Interpretation

The K-Means cluster plot reveals four distinct groupings of London boroughs that share similar patterns in building density and health-related characteristics.

Boroughs positioned close together within the same ellipse exhibit comparable socioeconomic and health profiles, while those further apart differ considerably.

The separation between clusters suggests meaningful variation in urban structure and lifestyle factors across London districts.

Overall, the K-Means clustering supports the earlier hierarchical analysis, confirming that

London boroughs can be meaningfully classified into four broad groups based on shared environmental and health attributes.

Matrix of Cluster Means (K-Means Method)

Cluster	Dom_Build	Smoking	NonDom_Build	Obese
1	0.274	0.256	0.141	0.523
2	0.705	0.579	0.492	0.350
3	0.262	0.639	0.199	0.733
4	0.404	0.545	0.758	0.117

Interpretation

The matrix of cluster means summarises the average standardized values of each variable within the four clusters derived from the K-Means algorithm.

Cluster 1 shows moderate residential density (Dom_Build = 0.27) and obesity levels, indicating relatively balanced urban and health conditions.

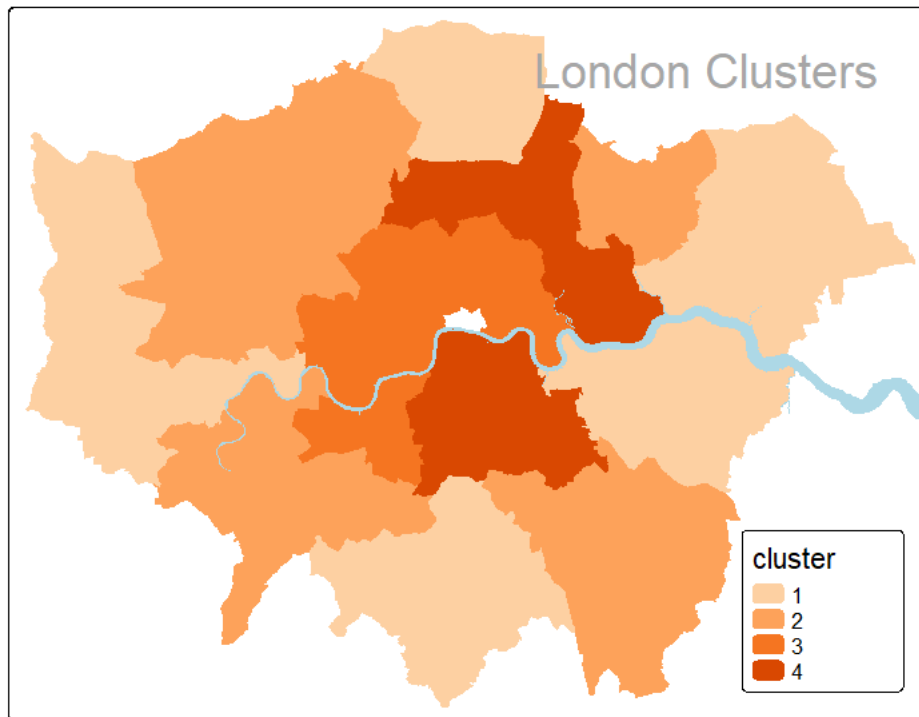
Cluster 2 has the highest mean values for Dom_Build and NonDom_Build, suggesting areas with the greatest urban density and commercial activity.

Cluster 3 records the highest Smoking and Obese means, representing boroughs with less favourable health outcomes.

Cluster 4 has the highest NonDom_Build but the lowest Obese score, reflecting more developed commercial areas with relatively healthier populations.

Overall, the results demonstrate clear differentiation between clusters, with urban structure and health indicators jointly defining the grouping of London boroughs.

Spatial Map of London Borough Clusters



Interpretation

The spatial map shows four distinct clusters of London boroughs based on urban and health characteristics.

Central areas form a cluster with high building density and less favorable health indicators, while outer boroughs group together with lower smoking and obesity levels and more open residential environments.

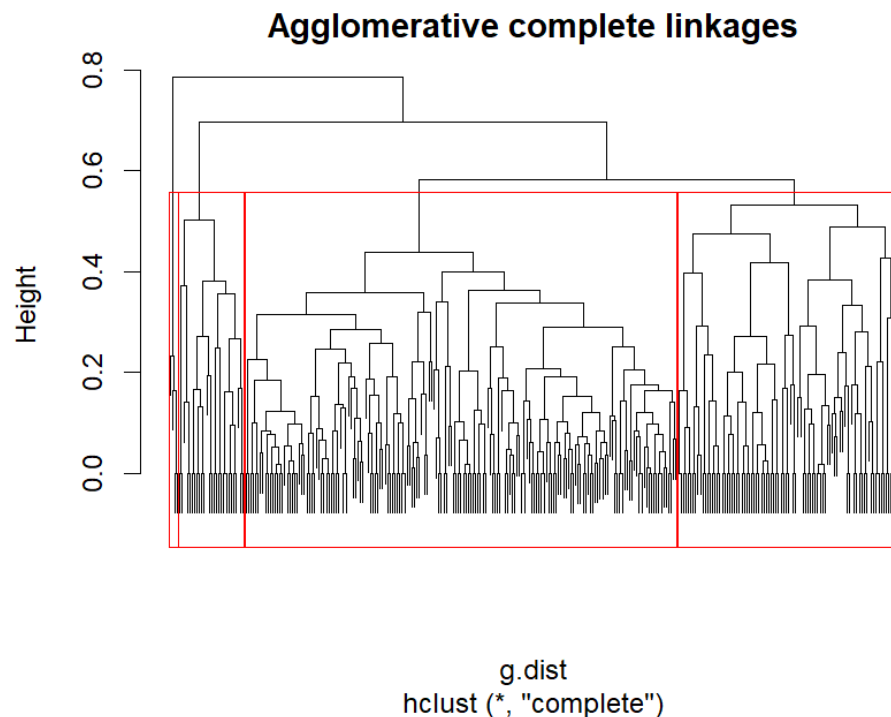
This spatial pattern mirrors the hierarchical clustering results, confirming a clear urban–suburban divide in socioeconomic and health conditions across London.

Comparison of Ward's and K-Means Clusters

Both Ward's and K-Means produced four broadly similar clusters, with central boroughs forming high-density, less healthy groups and outer boroughs showing lower density and better health. Minor differences appear in boundary areas for example, boroughs like Brent or Ealing shift cluster membership between the two methods, reflecting methodological differences between hierarchical and centroid-based clustering.

Overall, both methods highlight a consistent urban suburban divide in London's health and urban characteristics.

Dendrogram from the agglomerative clustering using Gowers dissimilarity.



Interpretation

The dendrogram based on Gower's dissimilarity and the complete linkage method reveals clear hierarchical groupings among the observations. Three main clusters can be identified, as indicated by the red boxes. The relatively low heights at which most clusters merge suggest that many observations are quite similar, while a few merge at higher dissimilarity levels, indicating greater diversity within those groups. Overall, the clustering captures distinct patterns in the data, reflecting differences in demographic, behavioral, or socioeconomic characteristics depending on the variables included.

Reflection

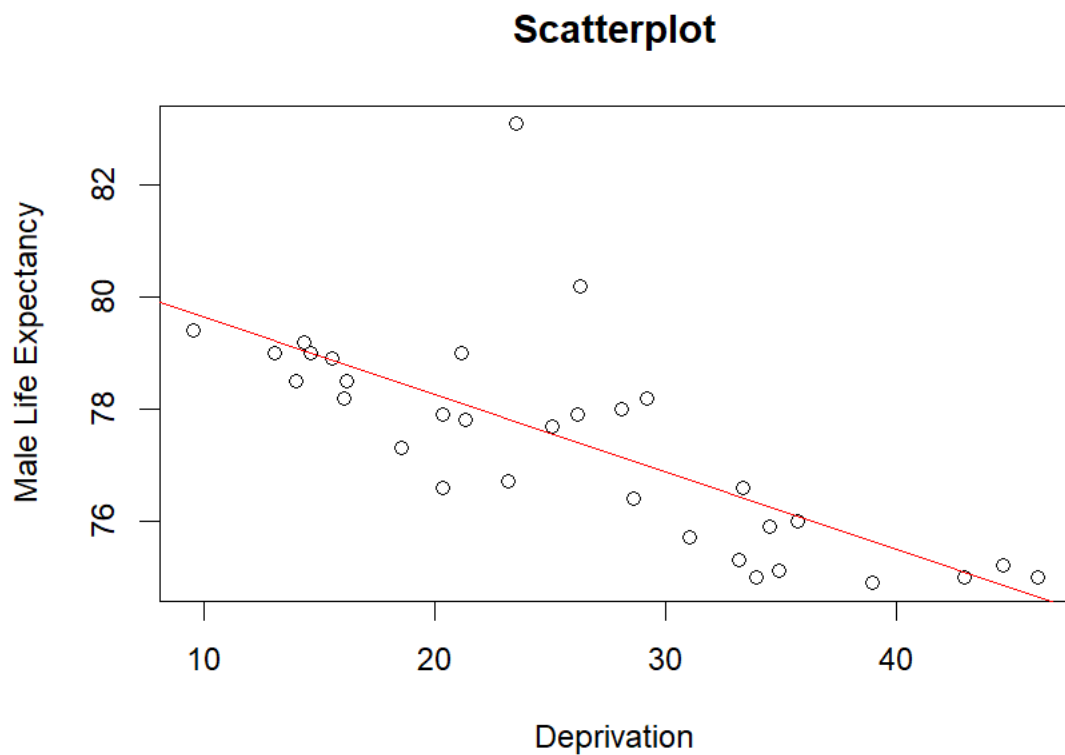
Throughout this session, I gained practical experience in using statistical and spatial analysis techniques in R. I learned how to perform correlation analysis, principal component analysis, and both hierarchical and k-means clustering. I also developed skills in data cleaning, visualisation,

and interpretation of complex multivariate relationships. Overall, this session enhanced my understanding of how data-driven methods can reveal meaningful patterns and support evidence-based decision-making.

Session 10

Regression Modelling

The regression line plot from Section 02



Interpretation

The scatterplot with regression line shows the relationship between Deprivation and Male Life Expectancy across London districts. A clear negative linear trend is visible which means higher deprivation is associated with lower male life expectancy.

Results

Spearman's rank correlation rho

```
data: Deprivation and Life_Male
S = 9843.8, p-value = 2.938e-08
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
-0.8042207
```

Interpretation

This result shows a strong negative association between deprivation and male life expectancy. As deprivation increases, male life expectancy decreases.

Output from the regression model

```
Call:
lm(formula = Life_Male ~ Deprivation)

Residuals:
    Min       1Q   Median       3Q      Max
-1.6092 -0.6776 -0.2439  0.2342  5.3304

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  81.02974   0.67302  120.398  < 2e-16 ***
Deprivation  -0.13867   0.02419   -5.733  2.95e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

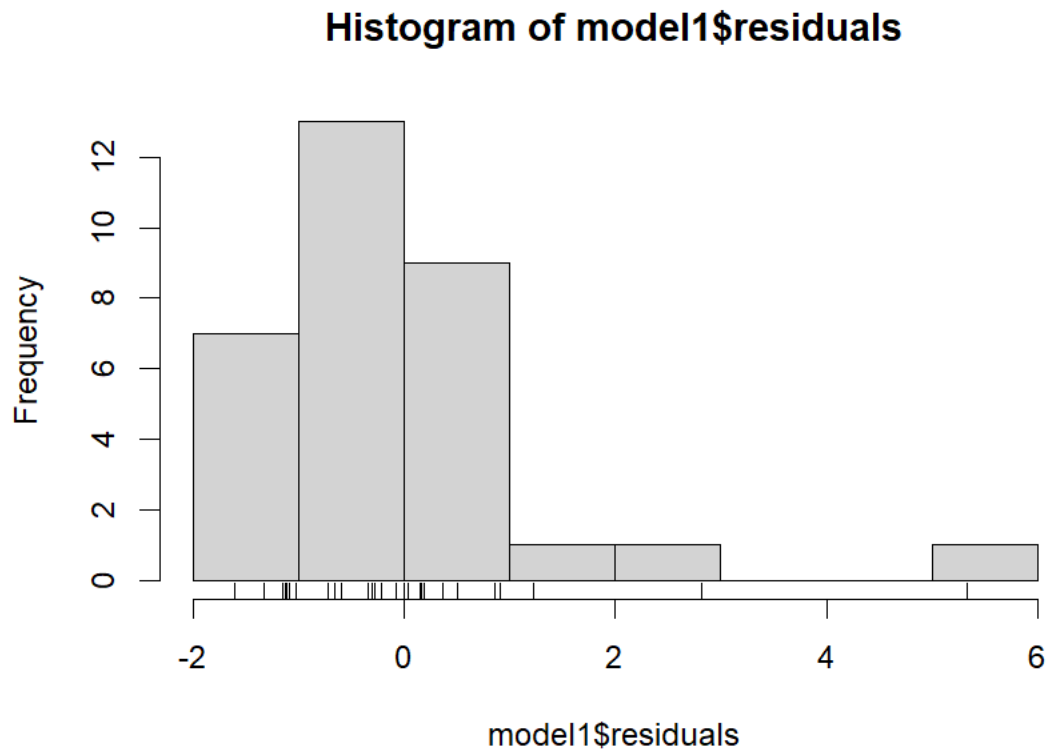
Residual standard error: 1.324 on 30 degrees of freedom
Multiple R-squared:  0.5228, Adjusted R-squared:  0.5069
F-statistic: 32.86 on 1 and 30 DF, p-value: 2.95e-06
```

Interpretation

Deprivation is a strong, statistically significant predictor of male life expectancy. For each 1-unit increase in deprivation, male life expectancy decreases by 0.14 years on average. The model explains 52.3% of the variance in life expectancy.

Plot of Residuals

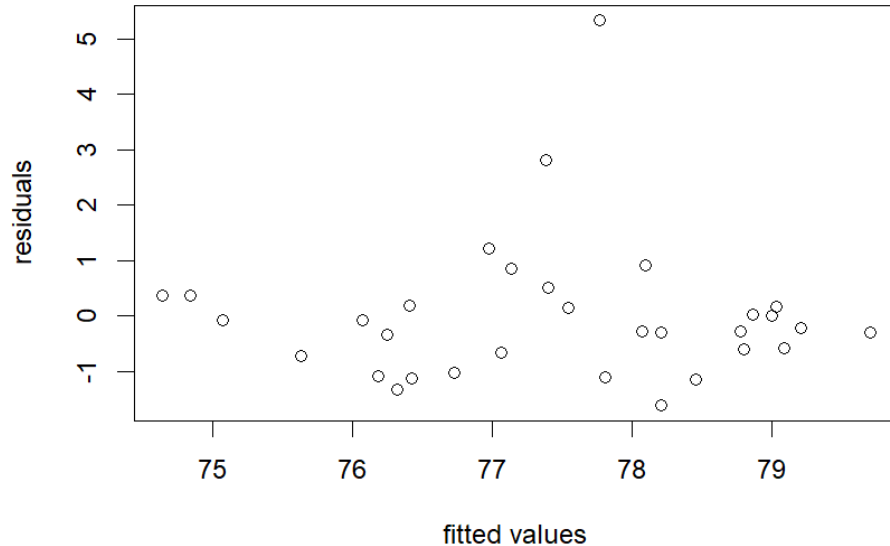
a) Residual Histogram



Interpretation

The histogram of residuals shows that the residuals are mostly centered around zero and roughly symmetric, indicating they are approximately normally distributed. Aside from one large positive outlier, there is no major skew, so the normality assumption is reasonably met.

b) Residuals vs Fitted Values Plot



Interpretation

The residuals are randomly scattered around zero with no clear pattern or funnel shape, indicating constant variance and suggesting that the linearity and homoscedasticity assumptions are reasonably satisfied.

Test of normality of the residuals

Result

Exact one-sample Kolmogorov-Smirnov test

data: model1\$residuals

D = 0.20279, p-value = 0.1248

alternative hypothesis: two-sided

Interpretation

Since $p > 0.05$, so we fail to reject the null hypothesis of normality. The residuals are consistent with a normal distribution, supporting regression assumptions.

Outputs from Section 4

Regression Summary for Model 2 (All Ten Predictors)

```
summary(model2)
```

```
Call:
```

```
lm(formula = Life_Male ~ Dom_Build + NonDom_Build + Dom_Gardens +  
    Greenspace + Smoking + Binge_Drink + Obese + Episodes + Benefits +  
    Crime)
```

```
Residuals:
```

```
      Min       1Q   Median       3Q      Max  
-2.02272 -0.44289  0.00305  0.70921  1.38197
```

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	84.9067214	5.5176720	15.388	6.57e-13	***
Dom_Build	0.2820230	0.1241387	2.272	0.0337	*
NonDom_Build	0.3465355	0.2185814	1.585	0.1278	
Dom_Gardens	-0.0553691	0.0534437	-1.036	0.3120	
Greenspace	0.0343311	0.0433610	0.792	0.4374	
Smoking	-0.0748256	0.0652076	-1.147	0.2641	
Binge_Drink	-0.4092845	0.2385194	-1.716	0.1009	
Obese	0.0928405	0.1143755	0.812	0.4261	
Episodes	0.0003746	0.0170236	0.022	0.9826	
Benefits	-0.0332028	0.0061391	-5.408	2.30e-05	***
Crime	-0.0084368	0.0110463	-0.764	0.4535	

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.9526 on 21 degrees of freedom
```

```
Multiple R-squared:  0.8272,    Adjusted R-squared:  0.7449
```

```
F-statistic: 10.05 on 10 and 21 DF,  p-value: 5.84e-06
```

Comments

Model 2 explains a large proportion of the variation in male life expectancy (high adjusted value), and the overall F-test is highly significant, indicating that the model fits the data well. However, only Dom_Build and especially Benefits are statistically significant predictors, while most other variables are not. Combined with the earlier VIF results showing high multicollinearity, this suggests that the model is over-complex and unstable, many predictors are overlapping, making individual coefficients hard to interpret.

Variance Inflation Factors (Model 2)

Variable	VIF
Dom_Build	5.83
NonDom_Build	23.19
Dom_Gardens	4.24
Greenspace	9.29
Smoking	2.14
Binge_Drink	7.08
Obese	3.90
Episodes	4.17
Benefits	3.49
Crime	6.41

Comments

The VIF results for Model 2 show that several predictors have high multicollinearity, especially NonDom_Build (23.19), Greenspace (9.29), Binge_Drink (7.08), and Crime (6.41). Many of these values exceed the usual threshold of 5, indicating that the independent variables are strongly correlated with each other. Although the model appears to explain a good amount of variation in male life expectancy, the high multicollinearity makes the regression coefficients unstable and difficult to interpret. This means Model 2 is not reliable in its current form and needs to be simplified by reducing or combining correlated variables.

VIF Check for Model 2

```
sqrt(vif(model2)) > 2 # if > 2 vif too high
```

Dom_Build	NonDom_Build	Dom_Gardens	Greenspace	Smoking	Binge_Drink
TRUE	TRUE	TRUE	TRUE	FALSE	TRUE
Obese	Episodes	Benefits	Crime		
FALSE	TRUE	FALSE	TRUE		

Comments

Several predictors in Model 2 have VIF values greater than 2 (shown as TRUE), including Dom_Build, NonDom_Build, Dom_Gardens, Greenspace, Binge_Drink, Episodes, and Crime. This indicates notable multicollinearity among these variables. Only Smoking, Obese, and Benefits are below the threshold. Overall, the model suffers from substantial collinearity, making coefficient estimates less stable and harder to interpret, and justifying the move to simplified models in later steps.

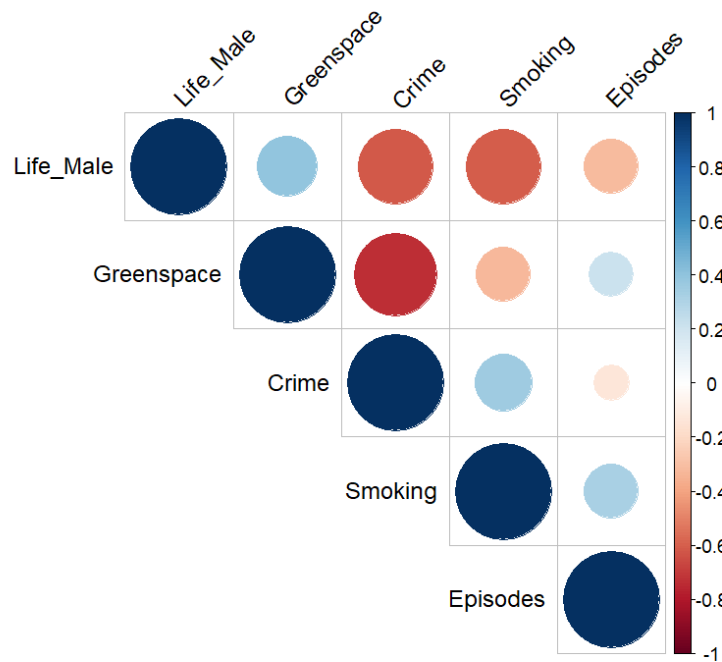
Correlation Matrix for Four Component Variables

Variable	Life_Male	Greenspace	Crime	Smoking	Episodes
Life_Male	1.00	0.40	-0.61	-0.61	-0.32
Greenspace	0.40	1.00	-0.74	-0.32	0.22
Crime	-0.61	-0.74	1.00	0.36	-0.14
Smoking	-0.61	-0.32	0.36	1.00	0.32
Episodes	-0.32	0.22	-0.14	0.32	1.00

Comments

The four component-related variables show moderate to low correlations with each other, indicating much less multicollinearity than in the full model. The strongest relationship is between Greenspace and Crime (-0.74), but all other pairs show weaker associations. Life_Male relates positively to Greenspace and negatively to Crime and Smoking, which is expected. Overall, this reduced set of predictors forms a more stable base for multiple regression.

Correlation Plot for Life_Male and Key Predictors



Comments

The plot shows that Life_Male is positively correlated with Greenspace and negatively correlated with Crime, Smoking, and Episodes, with the strongest negative links to Crime and Smoking. Greenspace and Crime are strongly negatively correlated with each other, while the other predictor–predictor correlations are more moderate. This confirms the expected directions of relationships and suggests some remaining collinearity, especially between Greenspace and Crime.

Summary(model3)

```
Call:
lm(formula = Life_Male ~ Greenspace + Crime + Smoking + Episodes)
```

Residuals:

```
   Min     1Q  Median     3Q      Max 
-2.6418 -0.7812 -0.1888  0.6973  3.3464
```

Coefficients:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 89.400085   3.020137  29.601 < 2e-16 ***
Greenspace   0.043878   0.025460   1.723  0.09625 .
Crime        -0.005213   0.008138  -0.641  0.52717
Smoking      -0.177092   0.074085  -2.390  0.02407 *
Episodes     -0.044568   0.014397  -3.096  0.00454 **
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.368 on 27 degrees of freedom

Multiple R-squared: 0.5415, Adjusted R-squared: 0.4736

F-statistic: 7.973 on 4 and 27 DF, p-value: 0.0002226

Comment

This model, using the four component-related variables, explains a moderate amount of variation in male life expectancy, with an adjusted value of around 0.47. Among the predictors, Smoking and Episodes are statistically significant, both showing negative effects on life expectancy. Greenspace is positive but only weakly significant, while Crime is not significant in this model. The residual standard error (1.37) is similar to the simple regression model, indicating reasonable fit. Overall, this reduced model performs better than the full model in terms of interpretability, but still includes variables with limited significance, suggesting that further refinement—such as removing Crime—could improve the model.

Output (VIF Check for Model 3)

```
> sqrt(vif(model3)) > 2
```

```
Greenspace  Crime  Smoking  Episodes
FALSE      FALSE  FALSE     FALSE
```

Comments

All VIF values for this model are below the threshold of 2, indicating that multicollinearity is not a concern among these four predictors. This confirms that the reduced set of variables forms a stable model where each predictor provides distinct information. Compared to the full model, this is a clear improvement, as the predictors are no longer strongly correlated with each other.

Calculate partial correlation

Output

```
> pcor.test(Life_Male, Crime, Greenspace)
  estimate p.value statistic n gp Method
1 -0.02933233 0.8755304 -0.1580274 32 1 pearson
> pcor.test(Life_Male, Greenspace, Crime)
  estimate p.value statistic n gp Method
1 0.21626 0.2426071 1.192823 32 1 pearson
> model3a <- lm(Life_Male ~ Greenspace + Smoking + Episodes)
```

Comments

The partial correlation results show that Crime has almost no independent relationship with male life expectancy after controlling for Greenspace (partial r approximately = -0.03 , $p = 0.88$). Likewise, Greenspace has only a weak and non-significant partial association with life expectancy when controlling Crime (partial r approximately = 0.22 , $p = 0.24$). These results indicate that Crime does not meaningfully contribute to explaining life expectancy once shared variance is accounted for. Therefore, removing Crime from the model is appropriate, leading to the updated model (model3a) using Greenspace, Smoking, and Episodes, which should improve model interpretability and reduce unnecessary predictors.

Summary(model3a)

output:

```
Call:
lm(formula = Life_Male ~ Greenspace + Smoking + Episodes)
```

Residuals:

```
   Min     1Q  Median     3Q      Max 
-2.6361 -0.7323 -0.2256  0.6953  3.5249
```

Coefficients:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 88.31182   2.47066  35.744 < 2e-16 ***
Greenspace   0.05122   0.02249   2.277  0.03059 *
Smoking     -0.18946   0.07077  -2.677  0.01227 *
Episodes     -0.04161   0.01349  -3.084  0.00456 **
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
Residual standard error: 1.354 on 28 degrees of freedom
Multiple R-squared:  0.5346, Adjusted R-squared:  0.4847
F-statistic: 10.72 on 3 and 28 DF, p-value: 7.298e-05
```

Comments:

This refined model shows improved interpretability, with all three predictors statistically significant. Greenspace has a positive association with male life expectancy, while both Smoking and Episodes have significant negative effects. The adjusted value (about 0.48) indicates that the model explains a moderate proportion of the variation in life expectancy, similar to the previous model but with fewer predictors. The residual standard error (1.35) is slightly lower, suggesting a slightly better fit. Overall, this model is more efficient and more reliable than earlier versions because all included predictors contribute meaningfully to the explanation.

Output (VIF Check for Model 3a)

```
sqrt(vif(model3a)) > 2
```

```
Greenspace  Smoking  Episodes
FALSE      FALSE    FALSE
```

Comments

All VIF values in Model 3a are below the threshold of 2, showing that multicollinearity is not an issue. This means the predictors do not overlap strongly with one another and each contributes uniquely to the model. Combined with the significance of all predictors, this confirms that Model 3a is a stable and reliable regression model.

Relative importance of variables

Output:

```
calc.relimp(model3a, type = c("lmg"), rela = TRUE)
```

```
Response variable: Life_Male
```

```
Total response variance: 3.556613
```

```
Analysis based on 32 observations
```

```
3 Regressors:
```

```
Greenspace Smoking Episodes
```

```
Proportion of variance explained by model: 53.46%
```

```
Metrics are normalized to sum to 100% (rela=TRUE).
```

```
Relative importance metrics:
```

	lmg
Greenspace	0.1687922
Smoking	0.4577703
Episodes	0.3734376

```
Average coefficients for different model sizes:
```

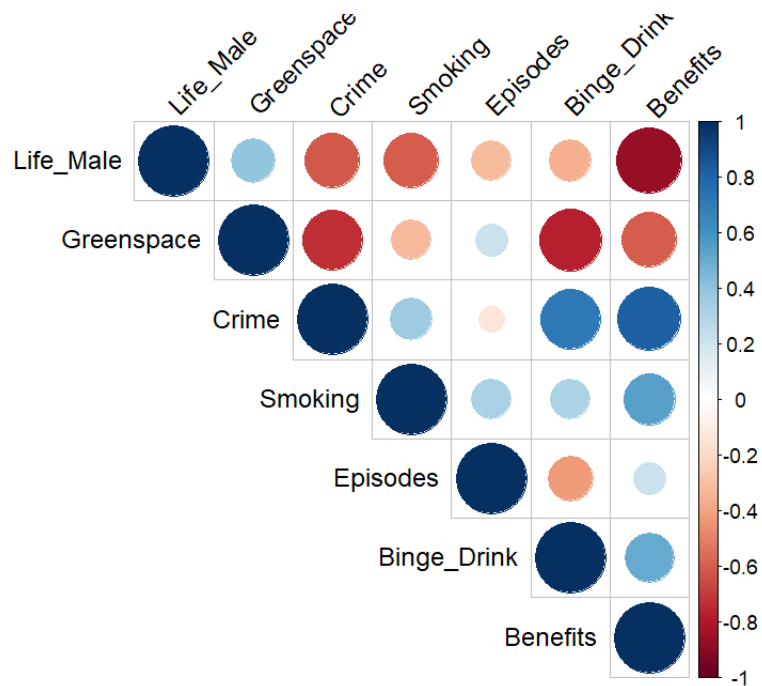
	1X	2Xs	3Xs
Greenspace	0.04367634	0.04799786	0.05122375
Smoking	-0.29247074	-0.25995405	-0.18946138
Episodes	-0.04354081	-0.04278427	-0.04161196

```
>
```

Comments

The relative importance results show that Smoking is the strongest predictor in the model, contributing about 46% to the explained variance in male life expectancy. Episodes is the next most influential predictor at around 37%, while Greenspace contributes about 17%. This indicates that behaviours and health-related workload factors play a larger role in predicting male life expectancy than environmental factors like greenspace. The importance values are consistent with the regression coefficients, further supporting the reliability of Model 3a.

Correlation Plot (Additional Variables Included)



Interpretation of Correlation Plot

The correlation plot shows several strong correlations among the variables added to the model. Greenspace and Crime display a strong negative relationship, while Binge_Drink and Benefits have strong positive correlations with each other and with Crime. In contrast, Life_Male shows negative associations with Crime, Smoking, and Binge_Drink, and a positive relationship with Greenspace. The presence of these strong intercorrelations indicates potential multicollinearity issues if all variables are included in a single regression model. This supports the need for careful variable selection or stepwise modelling to avoid instability in coefficient estimates.

Summary model3b

```
summary(model3b)
```

```
Call:
```

```
lm(formula = Life_Male ~ Greenspace + Crime + Smoking + Episodes +  
    Binge_Drink + Benefits)
```

```
Residuals:
```

```
      Min       1Q   Median       3Q      Max  
-1.8013 -0.5450 -0.1715  0.5892  3.0345
```

```
Coefficients:
```

```
              Estimate Std. Error t value Pr(>|t|)  
(Intercept) 88.933737   4.076275  21.817 < 2e-16 ***  
Greenspace  -0.024638   0.027122  -0.908 0.372325  
Crime         0.009846   0.007213   1.365 0.184414  
Smoking      -0.086204   0.064657  -1.333 0.194471  
Episodes     -0.015411   0.015032  -1.025 0.315071  
Binge_Drink -0.105941   0.191090  -0.554 0.584229  
Benefits     -0.028909   0.006606  -4.376 0.000188 ***
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 1.057 on 25 degrees of freedom
```

```
Multiple R-squared:  0.7465, Adjusted R-squared:  0.6857
```

```
F-statistic: 12.27 on 6 and 25 DF, p-value: 2.033e-06
```

Comments

Although this model explains a large portion of variation in male life expectancy, only Benefits is a significant predictor. All other variables become non-significant due to multicollinearity, meaning the model is unstable and not reliable for interpreting individual effects.

VIF Check for Model 3b

```
sqrt(vif(model3b)) > 2
```

Greenspace	Crime	Smoking	Episodes	Binge_Drink	Benefits
FALSE	FALSE	FALSE	FALSE	FALSE	FALSE

Comments

All VIF values are below 2, indicating no multicollinearity problems in this model despite most predictors being non-significant.

Output (Relative Importance Analysis for Model 3b)

```
calc.relimp(model3b, type = c("lmg"), rela = TRUE)
```

Response variable: Life_Male

Total response variance: 3.556613

Analysis based on 32 observations

6 Regressors:

Greenspace Crime Smoking Episodes Binge_Drink Benefits

Proportion of variance explained by model: 74.65%

Metrics are normalized to sum to 100% (rela=TRUE).

Relative importance metrics:

	lmg
Greenspace	0.04946814
Crime	0.03491201
Smoking	0.19045203
Episodes	0.18668103
Binge_Drink	0.03731938
Benefits	0.50116741

Average coefficients for different model sizes:

	1X	2Xs	3Xs	4Xs	5X
S					
Greenspace	0.043676336	0.025665763	0.010521388	-0.002358334	-0.01435077
9					
Crime	-0.008757773	-0.002106095	0.002117963	0.005003702	0.00747868
5					
Smoking	-0.292470735	-0.244427156	-0.185241125	-0.140948521	-0.10840423
6					
Episodes	-0.043540813	-0.047699205	-0.042989933	-0.034524530	-0.02477992
4					
Binge_Drink	-0.220726515	-0.122565028	-0.091488688	-0.089680357	-0.09822460
5					
Benefits	-0.028130038	-0.029901792	-0.029781397	-0.029137734	-0.02881753
2					
		6Xs			
Greenspace		-0.024638147			
Crime		0.009845977			
Smoking		-0.086204296			
Episodes		-0.015411036			
Binge_Drink		-0.105940750			
Benefits		-0.028909124			

Comments

Benefits is the dominant predictor, contributing about half of the explained variance. The other variables contribute much smaller shares, confirming that Benefits is the main driver in this model while the remaining predictors add relatively little explanatory value.

Output (Forward Stepwise Selection for Model 4)

```
# forward stepwise selection  
> model4 <- stepwise(model3b, direction = "forward")
```

```
Direction: forward  
Criterion: BIC
```

```
Start: AIC=43.05  
Life_Male ~ 1
```

	Df	Sum of Sq	RSS	AIC
+ Benefits	1	66.505	43.750	16.940
+ Smoking	1	38.985	71.270	32.555
+ Episodes	1	24.749	85.506	38.383
<none>			110.255	43.052
+ Greenspace	1	8.552	101.703	43.934
+ Binge_Drink	1	5.504	104.751	44.879
+ Crime	1	3.654	106.601	45.439

```
Step: AIC=16.94  
Life_Male ~ Benefits
```

	Df	Sum of Sq	RSS	AIC
+ Episodes	1	10.0926	33.658	12.013
+ Crime	1	8.1580	35.592	13.802
+ Greenspace	1	6.4214	37.329	15.326
+ Smoking	1	5.4153	38.335	16.177
<none>			43.750	16.940
+ Binge_Drink	1	3.4183	40.332	17.802

```
Step: AIC=12.01  
Life_Male ~ Benefits + Episodes
```

	Df	Sum of Sq	RSS	AIC
<none>			33.658	12.013
+ Smoking	1	2.78134	30.876	12.719
+ Crime	1	1.56585	32.092	13.955
+ Greenspace	1	0.82892	32.829	14.681
+ Binge_Drink	1	0.18017	33.477	15.307

Comments

The stepwise procedure identifies Benefits as the strongest initial predictor, reducing the model error the most. Adding Episodes provides the next meaningful improvement, after which no additional variables substantially reduce the AIC. This indicates that a simpler model with just Benefits and Episodes is more efficient and performs better than models including the other predictors.

Output (Regression Summary for Model 4: Benefits + Episodes)

```
summary(model4)
```

```
Call:
```

```
lm(formula = Life_Male ~ Benefits + Episodes)
```

```
Residuals:
```

```
      Min       1Q   Median       3Q      Max
-2.2674 -0.5097 -0.0773  0.4953  3.1776
```

```
Coefficients:
```

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  87.773485   1.862490  47.127  < 2e-16 ***
Benefits     -0.025528   0.003819  -6.684 2.49e-07 ***
Episodes     -0.028578   0.009691  -2.949 0.00624 **
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 1.077 on 29 degrees of freedom
```

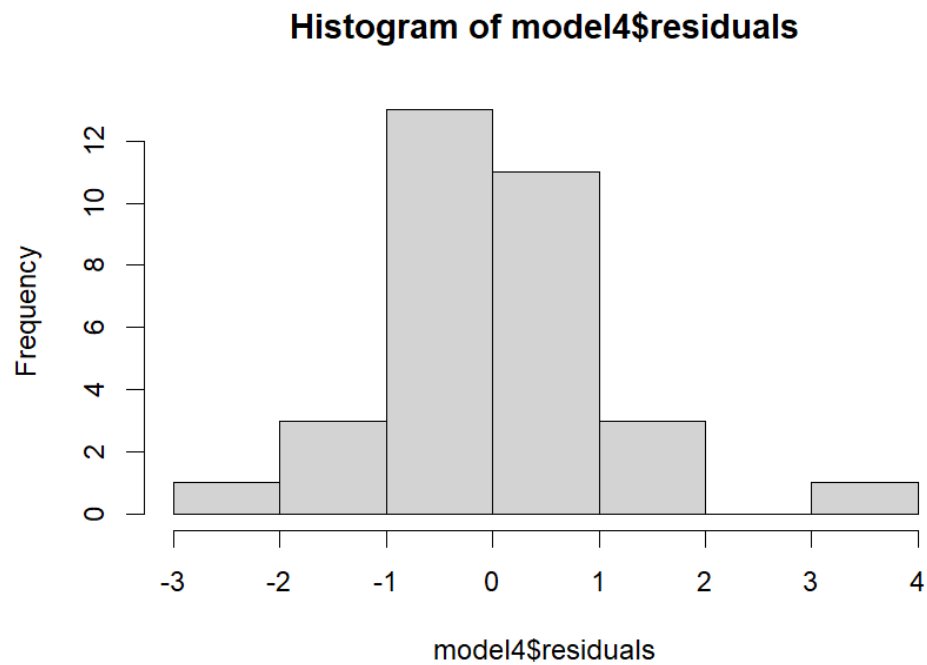
```
Multiple R-squared:  0.6947,    Adjusted R-squared:  0.6737
```

```
F-statistic:    33 on 2 and 29 DF,  p-value: 3.372e-08
```

Comments

This simplified model performs strongly, with both Benefits and Episodes emerging as significant negative predictors of male life expectancy. The model explains a substantial proportion of the variation, and the residual error is low, indicating a good overall fit.

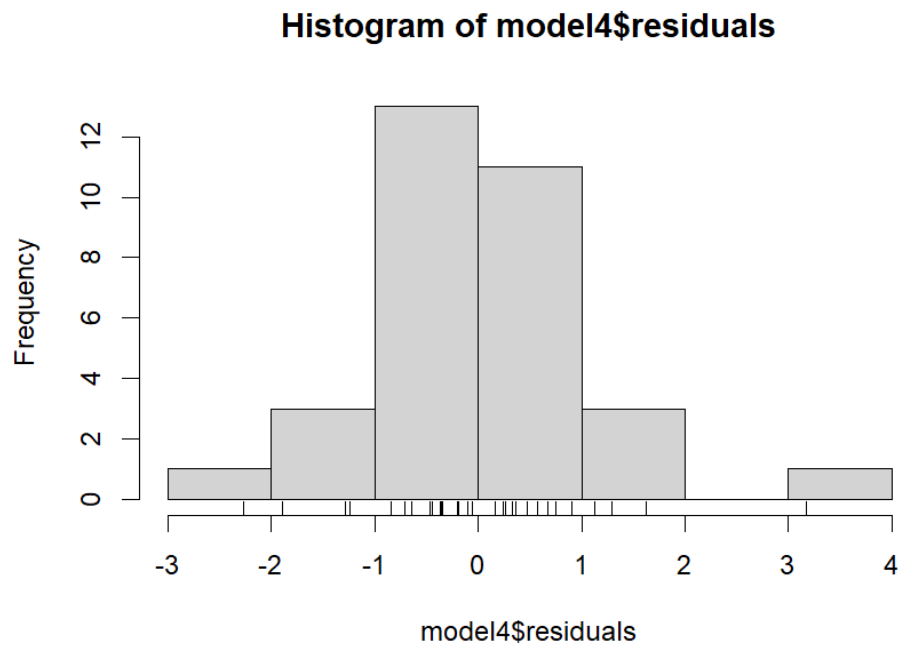
Histogram of Model 4 Residuals



Interpretation

The residuals are roughly centered around zero and show an approximately symmetric distribution, with only a few mild outliers. This suggests that the normality assumption is reasonably satisfied for Model 4.

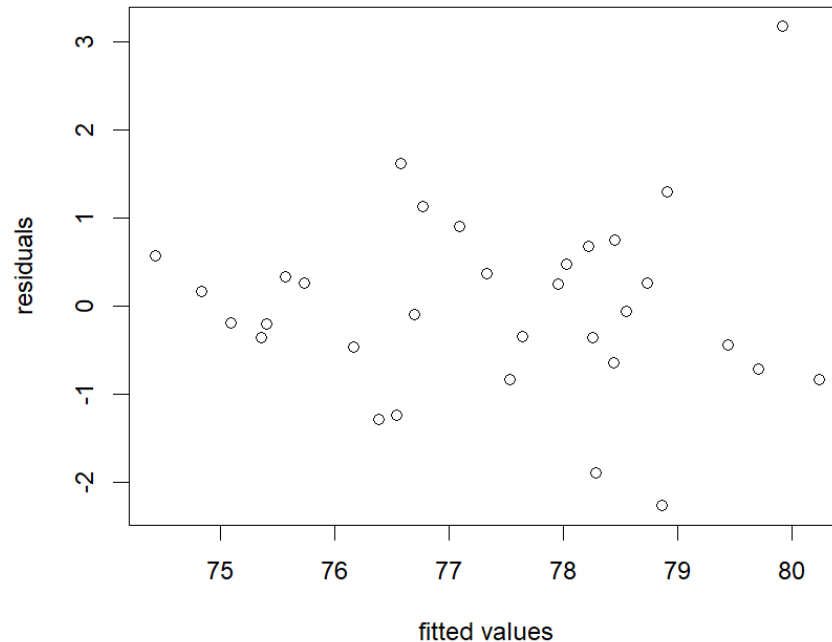
Output (Histogram and Rug Plot of Model 4 Residuals)



Interpretation

The residuals are centered around zero and appear roughly symmetric, with only a few mild outliers at both ends. This suggests that the normality assumption is reasonably met for Model 4, and the residual distribution does not indicate major problems.

Residuals vs Fitted Values Plot for Model 4



Interpretation

The residuals appear randomly scattered around zero with no visible pattern or funnel shape. This indicates that the assumptions of linearity and constant variance are reasonably satisfied, and there is no evidence of heteroscedasticity in Model 4.

Output (Kolmogorov–Smirnov Test for Model 4 Residuals)

```
Exact one-sample kolmogorov-smirnov test
```

```
data: model4$residuals  
D = 0.085178, p-value = 0.959  
alternative hypothesis: two-sided
```

Comments

The very high p-value (0.959) indicates no evidence against normality. This means the residuals follow a normal distribution closely, supporting the validity of Model 4's assumptions.

Output (VIF Check for Model 4)

```
sqrt(vif(model4)) > 2
Benefits Episodes
FALSE      FALSE
```

Interpretation

Both predictors have VIF values below 2, indicating no multicollinearity. This confirms that the variables in Model 4 are independent and the model is stable.

Output (Relative Importance Analysis for Model 4)

```
calc.relimp(model4, type = c("lmg"), rela = TRUE)
Response variable: Life_Male
Total response variance: 3.556613
Analysis based on 32 observations

2 Regressors:
Benefits Episodes
Proportion of variance explained by model: 69.47%
Metrics are normalized to sum to 100% (rela=TRUE).

Relative importance metrics:

                        lmg
Benefits 0.7725658
Episodes 0.2274342

Average coefficients for different model sizes:

                        1x          2xs
Benefits -0.02813004 -0.02552817
Episodes -0.04354081 -0.02857761
```

Comments

Benefits is the dominant predictor, accounting for about 77% of the explained variance, while Episodes contributes around 23%. This confirms that Benefits plays the strongest role in predicting male life expectancy within this simplified model.

Output (F-Test Comparing Model 2 and Model 4)

```
# test whether model2 and model4 are significantly different using F test
> anova(model2, model4, test = "F")
Analysis of Variance Table

Model 1: Life_Male ~ Dom_Build + NonDom_Build + Dom_Gardens + Greenspace +
  Smoking + Binge_Drink + Obese + Episodes + Benefits + Crime
Model 2: Life_Male ~ Benefits + Episodes
  Res.Df    RSS Df Sum of Sq    F Pr(>F)
1      21 19.055
2      29 33.658 -8    -14.602 2.0116 0.09536 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

>
```

Comments

The p-value of 0.095 indicates that the reduction from ten predictors in Model 2 to just two predictors in Model 4 does not lead to a statistically significant loss of model fit. Although Model 2 is larger and more complex, it does not provide a meaningful improvement over the simpler Model 4. This suggests that Benefits and Episodes capture most of the explanatory power, making Model 4 a more efficient and practical choice.

Output (Forward Stepwise Selection for Model 5)

```
# use a stepwise approach to search for a best model
> model5 <- stepwise(model2, direction = "forward")

Direction: forward
Criterion: BIC

Start: AIC=43.05
Life_Male ~ 1

      Df Sum of Sq    RSS    AIC
+ Benefits      1    66.505  43.750 16.940
+ Smoking       1    38.985  71.270 32.555
+ Episodes      1    24.749  85.506 38.383
<none>                 110.255 43.052
+ Greenspace    1     8.552 101.703 43.934
+ Obese         1     6.381 103.874 44.610
+ Binge_Drink   1     5.504 104.751 44.879
+ Crime         1     3.654 106.601 45.439
+ NonDom_Build  1     1.662 108.593 46.031
+ Dom_Gardens   1     0.939 109.316 46.244
+ Dom_Build     1     0.195 110.060 46.461
```

Step: AIC=16.94

Life_Male ~ Benefits

	Df	Sum of Sq	RSS	AIC
+ NonDom_Build	1	12.8190	30.931	9.3102
+ Episodes	1	10.0926	33.658	12.0133
+ Dom_Build	1	9.9109	33.839	12.1856
+ Crime	1	8.1580	35.592	13.8016
+ Greenspace	1	6.4214	37.329	15.3261
+ Smoking	1	5.4153	38.335	16.1772
<none>			43.750	16.9398
+ Obese	1	3.7548	39.995	17.5341
+ Dom_Gardens	1	3.6015	40.149	17.6565
+ Binge_Drink	1	3.4183	40.332	17.8022

Step: AIC=9.31

Life_Male ~ Benefits + NonDom_Build

	Df	Sum of Sq	RSS	AIC
+ Smoking	1	4.1399	26.791	8.1779
<none>			30.931	9.3102
+ Binge_Drink	1	2.3909	28.540	10.2015
+ Dom_Build	1	1.8398	29.091	10.8136
+ Episodes	1	0.4884	30.443	12.2666
+ Obese	1	0.4100	30.521	12.3489
+ Dom_Gardens	1	0.0954	30.836	12.6770
+ Greenspace	1	0.0150	30.916	12.7604
+ Crime	1	0.0062	30.925	12.7695

Step: AIC=8.18

Life_Male ~ Benefits + NonDom_Build + Smoking

	Df	Sum of Sq	RSS	AIC
<none>			26.791	8.1779
+ Obese	1	1.28754	25.504	10.0676
+ Binge_Drink	1	1.11453	25.677	10.2840
+ Dom_Build	1	0.94136	25.850	10.4991
+ Episodes	1	0.04493	26.746	11.5899
+ Dom_Gardens	1	0.01315	26.778	11.6279
+ Greenspace	1	0.01305	26.778	11.6281
+ Crime	1	0.00131	26.790	11.6421

Comments

The stepwise selection procedure identifies Benefits as the strongest initial predictor, followed by NonDom_Build and then Smoking as the variables that provide the greatest improvement in model fit. After these three predictors are included, adding any remaining variables produces only very small reductions in RSS and increases the AIC, indicating no meaningful improvement. This suggests that a more complex model offers little advantage, and the

combination of Benefits, NonDom_Build, and Smoking forms the most efficient model identified by the stepwise process.

Output (Regression Summary for Model 5: Benefits + NonDom_Build + Smoking)

```
summary(model5)
```

```
Call:
```

```
lm(formula = Life_Male ~ Benefits + NonDom_Build + Smoking)
```

```
Residuals:
```

```
      Min       1Q   Median       3Q      Max
-1.76509 -0.60252 -0.08724  0.47381  2.94708
```

```
Coefficients:
```

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  84.711561    1.092594   77.533  < 2e-16 ***
Benefits     -0.031172    0.004572   -6.818 2.09e-07 ***
NonDom_Build  0.191628    0.055171    3.473  0.00169 **
Smoking      -0.112156    0.053920   -2.080  0.04679 *
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.9782 on 28 degrees of freedom
```

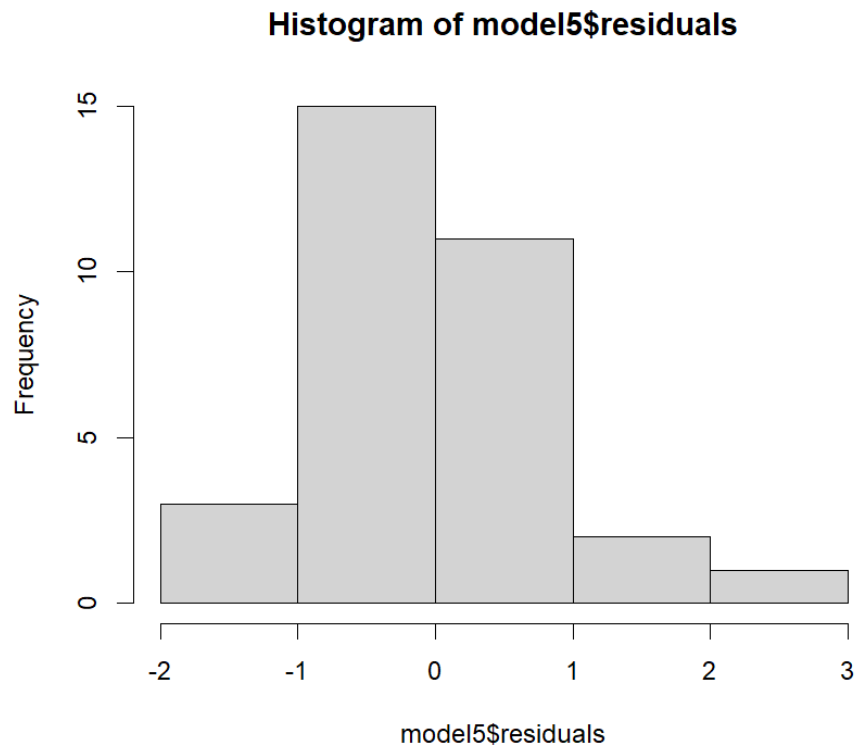
```
Multiple R-squared:  0.757,    Adjusted R-squared:  0.731
```

```
F-statistic: 29.08 on 3 and 28 DF,  p-value: 9.536e-09
```

Comments

This model explains a substantial proportion of variation in male life expectancy, with all three predictors statistically significant. Benefits remains the strongest predictor with a clear negative effect, while NonDom_Build shows a positive association, and Smoking shows a smaller negative effect. The residual standard error is low, and the adjusted value (about 0.73) indicates a strong overall fit. This model performs well statistically, but includes more predictors than necessary compared with simpler models that offer similar explanatory power.

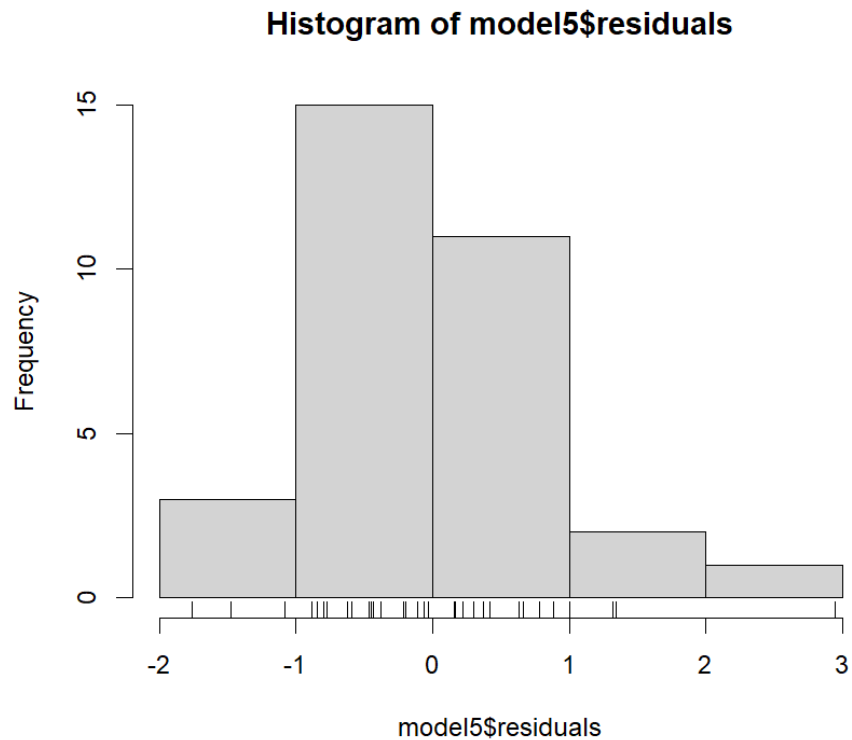
Histogram of Model 5 Residuals



Interpretation

The residuals appear roughly centred around zero with a fairly symmetric shape, suggesting no major deviations from normality. There are a few values extending slightly to the right, but overall the distribution looks acceptable, indicating that the normality assumption for the regression model is reasonably met.

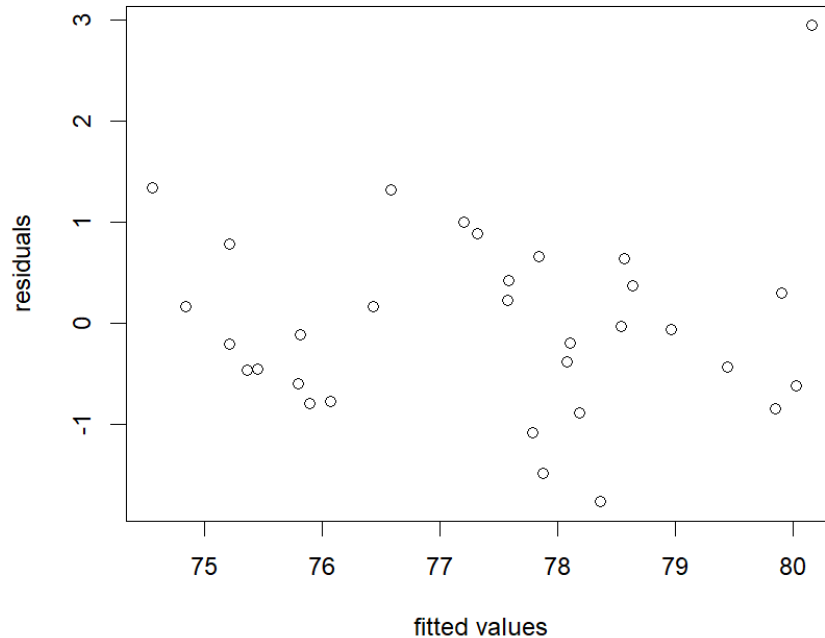
Histogram of Model 5 Residuals (with Rug Plot)



Interpretation

The residuals are centred near zero and show a reasonably symmetric distribution, with most values falling between -1 and $+1$. A few positive outliers extend the right tail slightly, but overall the pattern suggests that the normality assumption is largely acceptable for Model 5.

Residuals vs Fitted Plot (Model 5)



Comments

The residuals appear randomly scattered around zero with no visible pattern, suggesting that the assumptions of linearity and homoscedasticity are reasonably satisfied. There is no clear funnel shape or curvature, so the model's fit looks appropriate.

Kolmogorov–Smirnov Test for Normality (Model 5 Residuals)

Exact one-sample Kolmogorov-Smirnov test

data: model5\$residuals

D = 0.077749, p-value = 0.9821

alternative hypothesis: two-sided

Comments

The very high p-value (0.9821) indicates no evidence against normality. This means the residuals closely follow a normal distribution, supporting the validity of the regression model's normality assumption.

VIF Check for Model 5

```
sqrt(vif(model5)) > 2
Benefits NonDom_Build Smoking
FALSE FALSE FALSE
```

Comments

All VIF values are below 2, indicating no multicollinearity concerns among Benefits, NonDom_Build, and Smoking. This confirms that the predictors are not excessively correlated and are stable within the model.

Relative Importance Analysis (Model 5)

```
calc.relimp(model5, type = c("lmg"), rela = TRUE)
Response variable: Life_Male
Total response variance: 3.556613
Analysis based on 32 observations

3 Regressors:
Benefits NonDom_Build Smoking
Proportion of variance explained by model: 75.7%
Metrics are normalized to sum to 100% (rela=TRUE).

Relative importance metrics:

                                lmg
Benefits      0.66407435
NonDom_Build  0.07834407
Smoking       0.25758158

Average coefficients for different model sizes:

                                1X          2Xs          3Xs
Benefits      -0.02813004 -0.0295342 -0.03117245
NonDom_Build  -0.06142292  0.1021794  0.19162771
Smoking       -0.29247074 -0.2104809 -0.11215613

>
```

Comments

Benefits is by far the most influential predictor, contributing about 66% of the explained variance in life expectancy. Smoking accounts for roughly 26%, while NonDom_Build contributes only 8%. This confirms that the model's predictive power is driven mainly by Benefits, with Smoking also playing a meaningful but smaller role.

F-Test Comparing Model 2 and Model 5

```
# test whether model2 and model5 are significantly different using F test
> anova(model2, model5, test = "F")
Analysis of Variance Table

Model 1: Life_Male ~ Dom_Build + NonDom_Build + Dom_Gardens + Greenspace +
  Smoking + Binge_Drink + Obese + Episodes + Benefits + Crime
Model 2: Life_Male ~ Benefits + NonDom_Build + Smoking
  Res.Df    RSS Df Sum of Sq    F Pr(>F)
1      21 19.055
2      28 26.791 -7    -7.7362 1.218 0.3365

>
```

Comments

The p-value (0.3365) shows no significant difference between the full 10-predictor model (Model 2) and the simpler 3-predictor model (Model 5). This means removing the extra predictors does not significantly reduce model performance. Model 5 is therefore more efficient, offering similar explanatory power with far fewer variables.

Output (Regression Summary for Model 6: Factor Scores RC1–RC4)

```
> summary (model6)

Call:
lm(formula = Life_Male ~ fit.data$RC1 + fit.data$RC2 + fit.data$RC3 +
    fit.data$RC4)

Residuals:
    Min     1Q   Median     3Q     Max
-2.1736 -0.9078 -0.3432  0.7698  3.8454

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
```

```

(Intercept) 77.4125 0.2556 302.859 < 2e-16 ***
fit.data$RC1 -0.3586 0.2597 -1.381 0.17862
fit.data$RC2 -1.0562 0.2597 -4.067 0.00037 ***
fit.data$RC3 -0.2000 0.2597 -0.770 0.44786
fit.data$RC4 -0.6719 0.2597 -2.587 0.01538 *
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.446 on 27 degrees of freedom
Multiple R-squared: 0.488, Adjusted R-squared: 0.4122
F-statistic: 6.434 on 4 and 27 DF, p-value: 0.0009018

```

Comments

Model 6 explains a moderate proportion of variation in male life expectancy using the four factor-analysis components. Among these components, RC2 and RC4 are statistically significant negative predictors, while RC1 and RC3 are not significant. The model's adjusted value (about 0.41) is lower than previous regression models, and the residual error is slightly higher, indicating that the factor-based model does not perform as well. This suggests that using factor scores alone provides a weaker explanation of life expectancy compared with models built from the original variables.

Output (Forward Stepwise Selection for Model 7)

```

# use a stepwise approach to search for a best model
> model7 <- stepwise(model6, direction = "forward")

```

```

Direction: forward
Criterion: BIC

```

```

Start: AIC=43.05
Life_Male ~ 1

```

	Df	Sum of Sq	RSS	AIC
+ fit.data\$RC2	1	34.584	75.671	34.472
+ fit.data\$RC4	1	13.995	96.260	42.174
<none>			110.255	43.052
+ fit.data\$RC1	1	3.987	106.268	45.339
+ fit.data\$RC3	1	1.240	109.015	46.155

```

Step: AIC=34.47
Life_Male ~ fit.data$RC2

```

	Df	Sum of Sq	RSS	AIC
+ fit.data\$RC4	1	13.9948	61.676	31.394

```

<none>                                75.671 34.472
+ fit.data$RC1  1      3.9870 71.684 36.206
+ fit.data$RC3  1      1.2403 74.430 37.409

Step:  AIC=31.39
Life_Male ~ fit.data$RC2 + fit.data$RC4

              Df Sum of Sq    RSS   AIC
<none>                61.676 31.394
+ fit.data$RC1  1      3.9870 57.689 32.721
+ fit.data$RC3  1      1.2403 60.436 34.210
>

```

Comments

The stepwise procedure selects RC2 as the strongest initial predictor, followed by RC4, which further improves the model fit. Adding the remaining components (RC1 and RC3) does not substantially reduce the residual sum of squares or improve the AIC. This indicates that RC2 and RC4 capture the meaningful variation in life expectancy from the factor-based variables, while the other components add little explanatory value.

Output (Regression Summary for Model 7: RC2 + RC4)

```

summary(model7)

Call:
lm(formula = Life_Male ~ fit.data$RC2 + fit.data$RC4)

Residuals:
    Min       1Q   Median       3Q      Max
-2.4507 -1.1124 -0.0876  0.9234  3.2976

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   77.4125     0.2578 300.280 < 2e-16 ***
fit.data$RC2  -1.0562     0.2619  -4.033 0.000366 ***
fit.data$RC4  -0.6719     0.2619  -2.565 0.015748 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

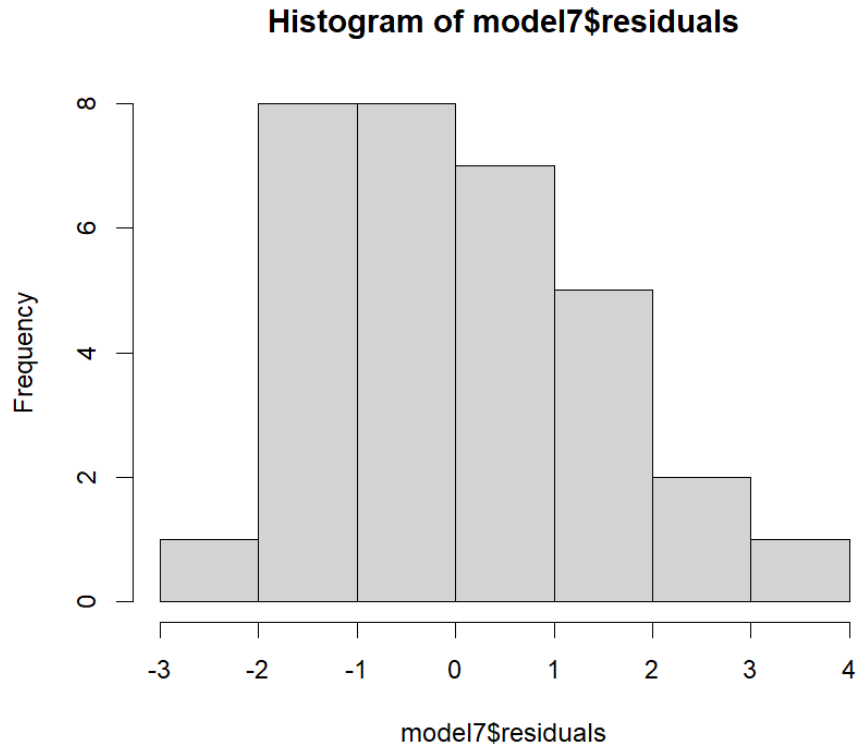
Residual standard error: 1.458 on 29 degrees of freedom
Multiple R-squared:  0.4406, Adjusted R-squared:  0.402
F-statistic: 11.42 on 2 and 29 DF, p-value: 0.0002197

```

Comments

This simplified factor-based model includes only the two significant components, RC2 and RC4, both of which show negative associations with male life expectancy. The adjusted value (about 0.40) indicates a moderate fit, though weaker than models based on the original variables. The residual error remains relatively high, suggesting that factor scores alone do not fully capture the key influences on life expectancy. Nonetheless, Model 7 is more efficient than Model 6, retaining only the meaningful components.

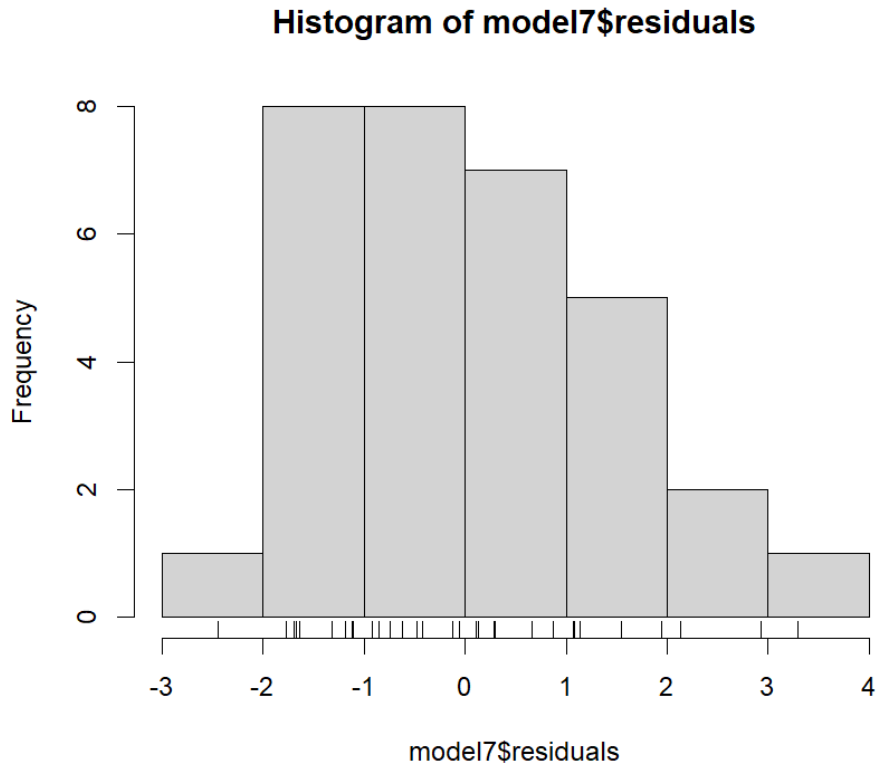
Histogram of Model 7 Residuals



Comments

The residuals show a roughly centered distribution but are slightly skewed, with a longer right tail and a few larger positive values. While the shape is not perfectly symmetric, it is still reasonably acceptable for regression. However, the deviations from normality are more noticeable here than in earlier models, reflecting the weaker fit of the factor-based model.

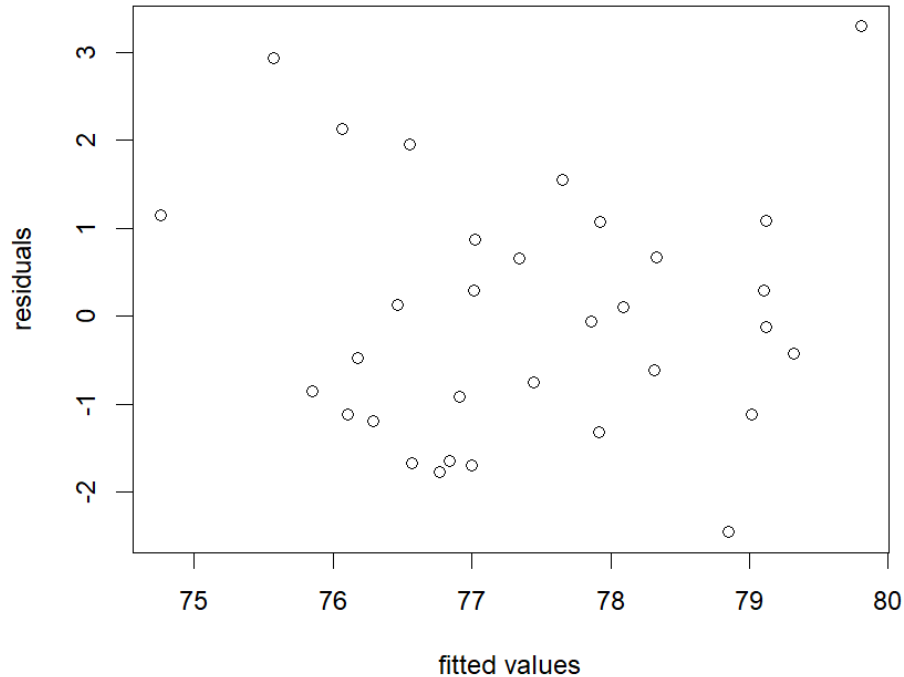
Histogram of Model 7 Residuals (with Rug Plot)



Comments

The residuals cluster around zero but show noticeable right-skewness, with several larger positive values extending the tail. Although the distribution is not severely distorted, it departs more from normality than earlier models, reflecting the weaker fit of the factor-based approach.

Residuals vs Fitted Plot (Model 7)



Comments

The residuals are scattered without a clear pattern, which is good, but the spread is wider and more uneven compared with earlier models. This suggests the linearity and homoscedasticity assumptions are only moderately satisfied. The model fit is weaker, consistent with the lower explanatory power of the factor-based predictors.

Kolmogorov–Smirnov Test for Normality (Model 7 Residuals)

Exact one-sample Kolmogorov-Smirnov test

data: model7\$residuals

D = 0.085792, p-value = 0.9565

alternative hypothesis: two-sided

Comments

The high p-value (0.9565) indicates no evidence against normality, meaning the residuals are consistent with a normal distribution. Despite the weaker overall model fit, the normality assumption is still adequately satisfied.

VIF Check for Model 7

```
sqrt(vif(model7)) > 2
fit.data$RC2 fit.data$RC4
FALSE FALSE
```

Comments

Both predictors (RC2 and RC4) show VIF values below 2, indicating no multicollinearity concerns. This means the two factor components included in Model 7 are not strongly correlated with each other and can be interpreted reliably within the regression model.

Relative Importance Analysis (Model 7)

```
> calc.relimp(model7, type = c("lmg"), rela = TRUE)
Response variable: Life_Male
Total response variance: 3.556613
Analysis based on 32 observations

2 Regressors:
fit.data$RC2 fit.data$RC4
Proportion of variance explained by model: 44.06%
Metrics are normalized to sum to 100% (rela=TRUE).

Relative importance metrics:

              lmg
fit.data$RC2 0.7119175
fit.data$RC4 0.2880825

Average coefficients for different model sizes:

              1X              2Xs
fit.data$RC2 -1.056230 -1.056230
fit.data$RC4 -0.671896 -0.671896
```

Comments

RC2 is the dominant predictor, contributing about 71% of the explained variance in life expectancy, while RC4 contributes the remaining 29%. The overall explanatory power is modest (about 44%), confirming that factor-based predictors capture less variation than models using the original variables.

F-Test Comparing Model 6 and Model 7

```
# test whether model7 and model6 are significantly different using F test
> anova(model6, model7, test = "F")
Analysis of Variance Table

Model 1: Life_Male ~ fit.data$RC1 + fit.data$RC2 + fit.data$RC3 + fit.data$RC4
Model 2: Life_Male ~ fit.data$RC2 + fit.data$RC4
  Res.Df    RSS Df Sum of Sq    F Pr(>F)
1      27 56.449
2      29 61.676 -2    -5.2273 1.2501 0.3025
```

Comments

The p-value (0.3025) shows no significant difference between the full factor-based model (Model 6) and the reduced model (Model 7). This means removing RC1 and RC3 does not worsen the model in a statistically meaningful way. RC2 and RC4 capture the relevant information effectively, making Model 7 a more efficient and equally adequate factor-based model.

Task 3:

Correlation Matrix for Low Birth Weight Dataset

Output

```
round(low.bwt.cor$correlations, 2)
      birth smoke ethnic m_age  mwt low_bwt
birth   1.00  0.03   0.00  0.64  0.30   0.09
smoke   0.03  1.00  -0.53 -0.04 -0.08   0.30
ethnic  0.00 -0.53   1.00 -0.15 -0.15  -0.32
m_age   0.64 -0.04  -0.15  1.00  0.32   0.09
mwt     0.30 -0.08  -0.15  0.32  1.00  -0.13
low_bwt 0.09  0.30  -0.32  0.09 -0.13   1.00

>
```

Comments

The matrix shows that smoking has a positive correlation with low birth weight (0.30), while ethnic group has a moderate negative correlation with low birth weight (−0.32), suggesting both may be important risk factors. Maternal age and weight have only weak correlations with low birth weight, indicating a smaller direct association in this sample.

Summary(mylogit2)

```
> summary(mylogit2)

Call:
glm(formula = low_bwt ~ smoke + ethnic + mwt, family = "binomial",
    data = low.bwt)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.329802   0.544486   0.606 0.544706
smoke1       0.516914   0.218877   2.362 0.018193 *
ethnic2     -0.784194   0.325853  -2.407 0.016102 *
ethnic3     -0.845182   0.251175  -3.365 0.000766 ***
mwt         -0.006939   0.003466  -2.002 0.045300 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 603.79  on 487  degrees of freedom
Residual deviance: 568.98  on 483  degrees of freedom
AIC: 578.98

Number of Fisher Scoring iterations: 4
```

Comments

This model shows that smoking, ethnicity, and maternal weight (mwt) are all statistically significant predictors of low birth weight. Smoking increases the odds of low birth weight (positive coefficient), while belonging to ethnic group 2 or 3 and having higher maternal weight both reduce the odds (negative coefficients). The reduction in deviance from 603.79 to 568.98 indicates the model fits the data better than the null model, and the AIC of 578.98 suggests a reasonable overall fit.

VIF Check for Logistic Regression Model (mylogit2)

```
> sqrt(vif(mylogit2)) > 2
      GVIF    Df GVIF^(1/(2*Df))
smoke FALSE FALSE             FALSE
ethnic FALSE FALSE             FALSE
mwt    FALSE FALSE             FALSE
>
```

Comments

All values are FALSE, meaning the VIFs for smoke, ethnic, and mwt are below the threshold for concern. This indicates there is no evidence of multicollinearity between these predictors, so the logistic regression coefficients can be interpreted with confidence.

Odds Ratios for Logistic Regression Model (mylogit2)

```
> # calculate Odds Ratio - Exp(b)
> exp(coef(mylogit2))
(Intercept)      smoke1      ethnic2      ethnic3      mwt
  1.3906929    1.6768442    0.4564874    0.4294791    0.9930852
```

Interpretation

- Smoking (smoke1, OR = 1.68): Babies of mothers who smoke have about 1.7 times higher odds of low birth weight than babies of non-smokers, controlling for ethnicity and maternal weight.
- Ethnicity (ethnic2, OR = 0.46; ethnic3, OR = 0.43): Compared with the reference ethnic group, babies in ethnic group 2 or 3 have lower odds of low birth weight (about 54% and 57% lower, respectively).
- Maternal weight (mwt, OR = 0.99): Higher maternal weight is associated with a small reduction in the odds of low birth weight; each unit increase in weight slightly decreases risk.

Conclusion about Risk Factors in Low Birth Weight

The analysis shows that maternal smoking is an important risk factor for low birth weight, significantly increasing the odds that a baby is born underweight. In contrast, babies from ethnic group 2 and ethnic group 3 have substantially lower odds of low birth weight compared with the reference ethnic group, suggesting a protective effect. Other variables such as birth order, maternal age, and maternal weight either become non-significant or have only small effects once smoking and ethnicity are included, so the main independent risk factor identified in this dataset is maternal smoking, with ethnicity also playing a strong role.

Conclusions for Section 10

The mean of the murder counts is 3.75 and the variance is 6.77, which are not equal. This indicates some over-dispersion. The quasi-Poisson model supports this, with a dispersion parameter slightly above 1, so it is reasonable to fit both Poisson and quasi-Poisson models.

In the full models (mypois1 and mypoiss2), Benefits, Greenspace, and NonDom_Build emerge as the main predictors, while the other variables are not significant. After refining the model to include only Benefits, Greenspace, and NonDom_Build (mypois3), all three remain important:

- Benefits has a positive effect, meaning higher levels of benefits are associated with slightly higher murder counts.
- Greenspace and NonDom_Build both have odds ratios below 1, suggesting that more greenspace and a higher proportion of non-domestic buildings are associated with lower murder counts, although the effect of greenspace is only borderline significant.

Relative importance measures and McFadden's R squared (about 0.22) indicate that the refined model has a modest but meaningful explanatory power, with Benefits being the strongest determinant, followed by NonDom_Build and then Greenspace. Overall, Section 10 shows that socio-economic conditions (benefits), urban form (non-domestic buildings), and environmental factors (greenspace) are key determinants of murder counts, and that a simple Poisson model with these three predictors provides a reasonable fit to the data.